

AIMS Lab

Advanced Intelligent Multidisciplinary Systems Lab

UNITED INTERNATIONAL UNIVERSITY

QUEST FOR EXCELLENCE

Prof. Khondaker A. Mamun, PhD

PhD. (Biomedical Engineering), University of Southampton, UK

Research Fellow (Biomedical Engineering), University of Toronto, Canada

Founder, CMED Health, A Digital Healthcare Platform

Founder and Director, Advanced Intelligent Multidisciplinary Systems Lab (AIMS Lab)

Professor, Dept. of CSE, United International University, Dhaka, Bangladesh



AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab

Innovation and Technological Adaptation for Healthcare and Disabilities

- Establishing Equitable, Preventive and Primary healthcare
- Health Inclusion, Referrals, Risk Prediction and Health Systems
- Access communication & Assistive Technologies for disables
- Human-Centered Design
- Human Computer Interaction (HCI)
- Industry 4.0 Technology - Internet of Things, Artificial Intelligence, Robotics
- Brain Computer Interface (BCI) Applications
- Neuroscience & Neural Engineering
- Data Science and Health Informatics
- Mobile Technology for Healthcare Applications

AIMS Lab

Advanced Intelligent Multidisciplinary Systems Lab







Digital Health

Digital Health Inclusion Towards Achieving Universal Health Coverage for Bangladesh

CMED

Connecting People, Saving Lives

Current Healthcare Situation in Bangladesh







bdnews24.com News Classifieds Business Opinion Sport Cricket Tube

Bangladesh can treat 80 percent patients at lower facilities: WHO expert

Senior Correspondent, bdnews24.com
Published: 04 Apr 2019 11:29 PM BdST | **Updated:** 04 Apr 2019 11:29 PM BdST



Without good health we can't retain this smile

Healthcare Situation in Bangladesh

2.5% of world population living in 0.03% Area





67% Death Due to NCD
Premature Death & Low Productivity

No Referral System Health Record

74% Out of Pocket Expense
No Health Protection/Insurance

$10B+ Spent on Healthcare/year
22.5% Patients Growth/Year

Research

AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab

CMED
Connecting People, Saving Lives

Research on Healthcare Situation in Bangladesh (2015)

ELSEVIER

Contents lists available at ScienceDirect

Computers and Electrical Engineering

journal homepage: www.elsevier.com/locate/compeleceng





CMED: Cloud based medical system framework for rural health monitoring in developing countries

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[b] Department of Computer Engineering, College of Computer and Information Sciences, King Saud University, Riyadh 11543, Saudi Arabia
[c] Department of Computer Science and Engineering, Bangladesh University of Business and Technology, Dhaka, Bangladesh
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[e] Advanced Intelligent Multidisciplinary Systems Lab (AIMS Lab), Department of Computer Science and Engineering, United International University, Dhaka, Bangladesh

AI4H Mobile applications & Cloud-based AI applications

Primary care
CD
NCD
MNCH
Family Planning
Mental Health
……..

ADA, Your MD, Babylon, CMED

Device + APP



Self

People

Risk Assessment
Design Thinking
Functionality
Contents
Clinical Accuracy
Health Record
Security/Privacy
……..

Health Agent

Health Volunteer

Last Mile Health Service CHW / HA

Dashboard



Cloud

AI
Accessibility
Cloud/Recovery
Health Record
Security/Privacy
Gov. Policy
……..

Policy Maker

Government

Referral Chain

Hospitals

Referral Chain

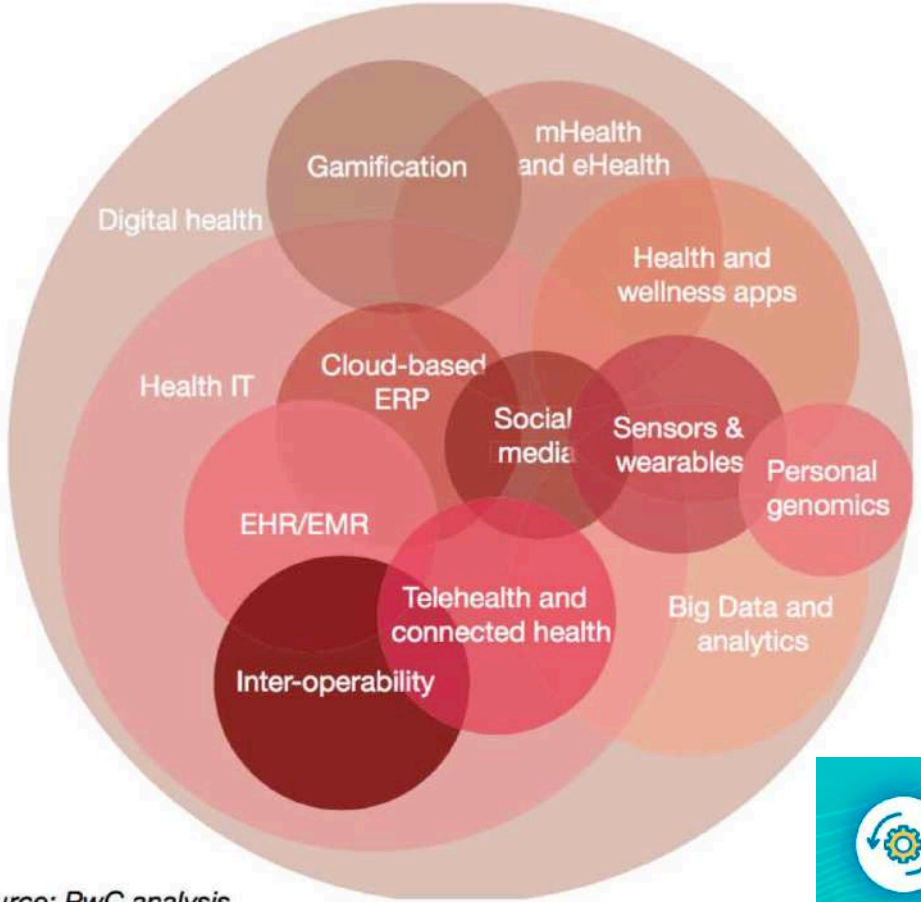
Doctors

Training & Adaptation

Digital Health Inclusion to achieve UHC

Digital health ecosystem

Digital health components



Source: PwC analysis

What does it mean for healthcare:

- Improve access
- Quality care
- Better information
- Better patient outcomes
- Increase patient engagement
- Enhance information flow



How CMED Digital Primary Healthcare Service Model Works



The Future of Healthcare

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Advanced Intelligent Multidisciplinary Systems Lab



Devi Prasad Shetty, MS, FRCS, is an Indian cardiac surgeon and entrepreneur. He is the chairman and founder of Narayana Health, a chain of 21 medical centers in India. He has performed more than 15,000 heart operations.



> Software will make smarter diagnoses than doctors in the next five years, and it will become legally mandatory for a doctor to take a second opinion from software before starting treatment. You need the right data, that's all.

CMED

Connecting People, Saving Lives

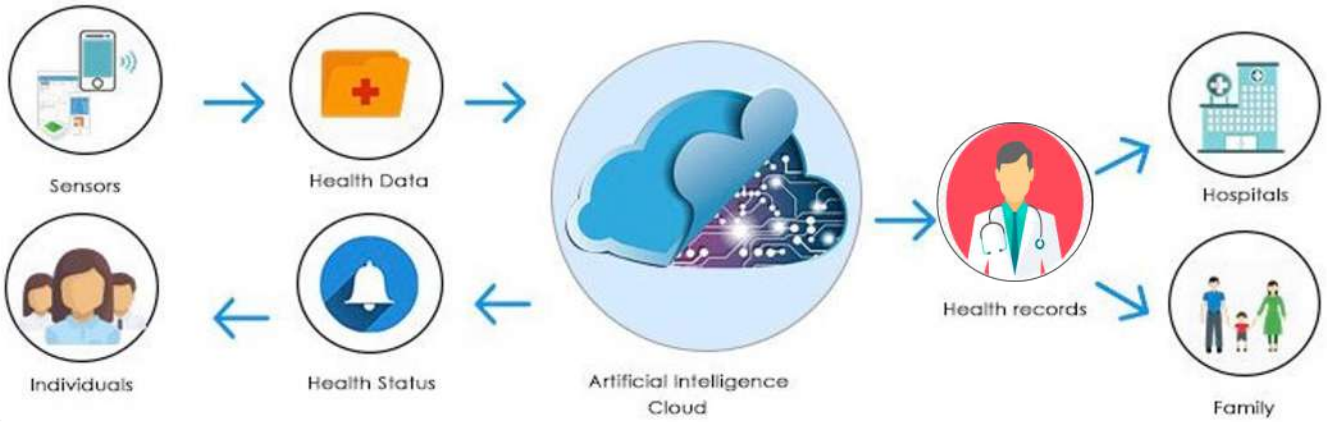
Last Mile Health Service for Universal Health Coverage



CMED Health Solution

Digital Healthcare Platform

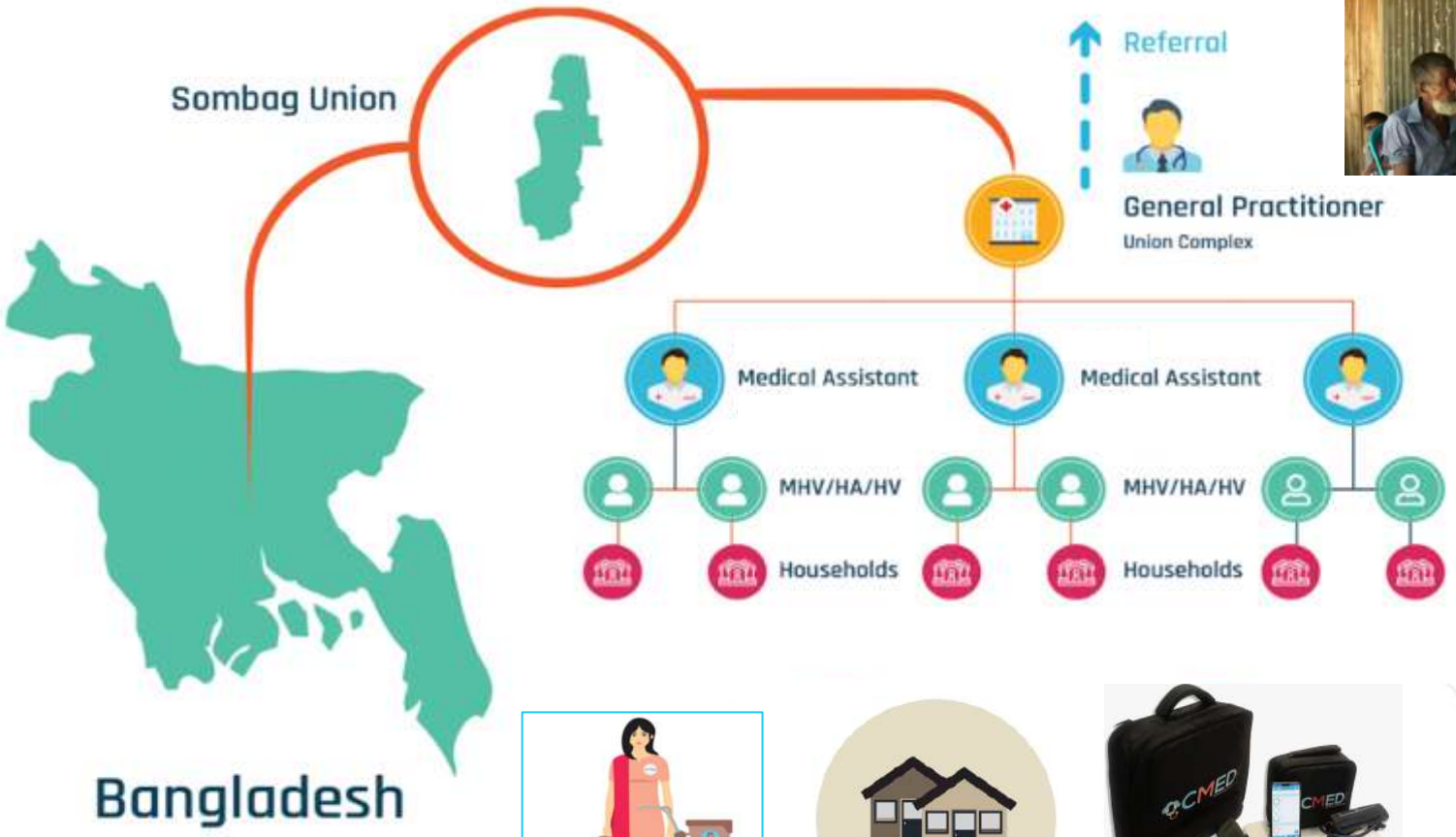
IoT enabled Artificial Intelligence Driven Cloud based System





- Health Monitoring
- Education and Awareness
- Risk Assessment & Referral
- Health Record & Ubiquitous Access to data
- AI Driven Risk Prediction for Stroke, Heart Attack and Diabetes

CMED Digital Healthcare Platform for Primary and Preventive Care





Doctor Consultation



Doorstep Healthcare Service & Referral



Health Visitor



Household



Smart Health Kit



Health Screening

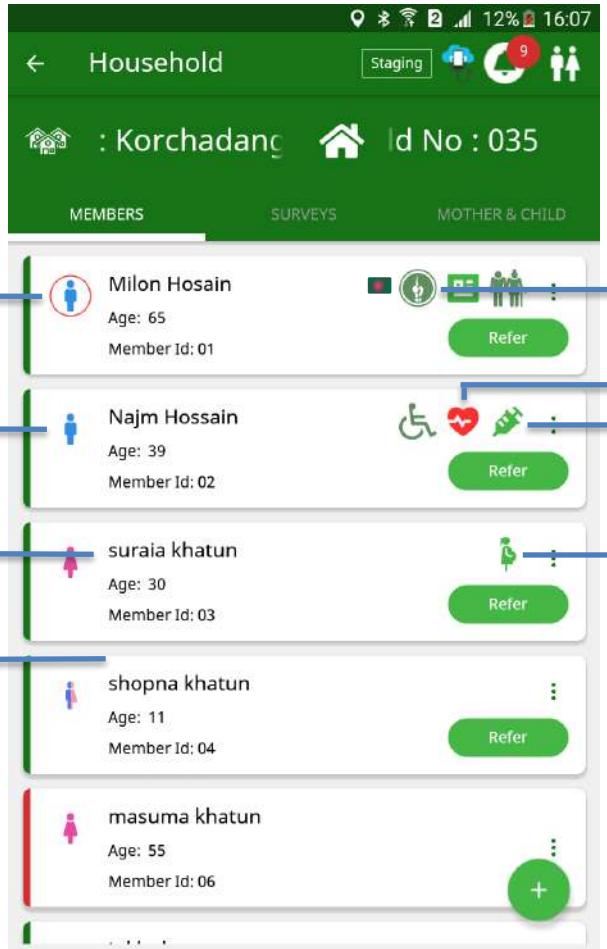
পল্লী কর্ম-সহায়ক ফাউন্ডেশন

CMED
Connecting People, Saving Lives

How CMED Digital Primary Healthcare Service Model Works



Benefits



Khana Prodhan

Gender info

Name

Dead

Freedom Fighter

Hypertension

Diabetic

Pregnant (Tracking ANC, PNC and other Services)

More icons can be added to identify new groups

CMED Digital Healthcare Platform for Primary and Preventive Care

E Health Record System

Name: **Habib Test** | Age: **24** | Gender: **Male** | CONDITIONS:
Contact No: **01648468446** | BLOOD GROUP : **B+** | REFERRED : **CMED CC** | CVD RISK STATUS:

PROFILE | MEASUREMENT | CVD RISK ASSESSMENT | PRESCRIPTIONS | LAB TEST RESULTS | REFERRED HISTORY

DISPLAYING PATIENTS PROFILE

FAMILY HISTORY

DIABETES	YES
HYPERTENSION	YES
HEART DISEASE	YES
STROKE	YES
C.O.P.D	YES
ASTHMA	YES
C.K.D	YES
CANCER	YES

PATIENTS HABIT

SMOKER	NO
TAKES CHEWABLE TOBACCO	NO

PREVIOUS MEDICAL HISTORY

DISEASE	STATUS	DURATION	DRUGS/ MEDICATION/ PREVENTION STEPS
DIABETES	YES	1 YEARS 1 MONTHS	Amlodipine 5 mg + Banzapril Hydrochloride 10 mg, Amlodipine 5 mg + Atenolol 25 mg,
HYPERTENSION	YES	1 YEARS 1 MONTHS	
HEART DISEASE	YES		
T.I.A	YES		
C.O.P.D	YES		
C.K.D	YES		



yasin Arafat
MBBS (Dhaka Medical College (DMC))
Dr., Bangladesh Medical

Name: Moin Chowdhury | **Age:** 32 | **Gender:** Male
PID: 01912500617 | **Blood Group:** B+ | **Date:**

Observation:
Pulse Rate: 74

Special Notes/Findings:
→ lack of sleep

Chief Complaint:
→ Abdominal Pain
→ Weakness
→ Muscle Pain (Myalgia)

Investigations:
→ Blood Culture

Provisional Diagnoses:
→ Typhoid fever, unspecified

℞
Tablet **Napa EXTRA**
Tablet **Histacin**

Advice:

Next visit:

চেম্বার:
CMED Health Clinic
Dhanmondi Satmasjid Road, Apt# C-5, House# 761, Safura Green, Dhaka 1209
Contact No: 01619195597

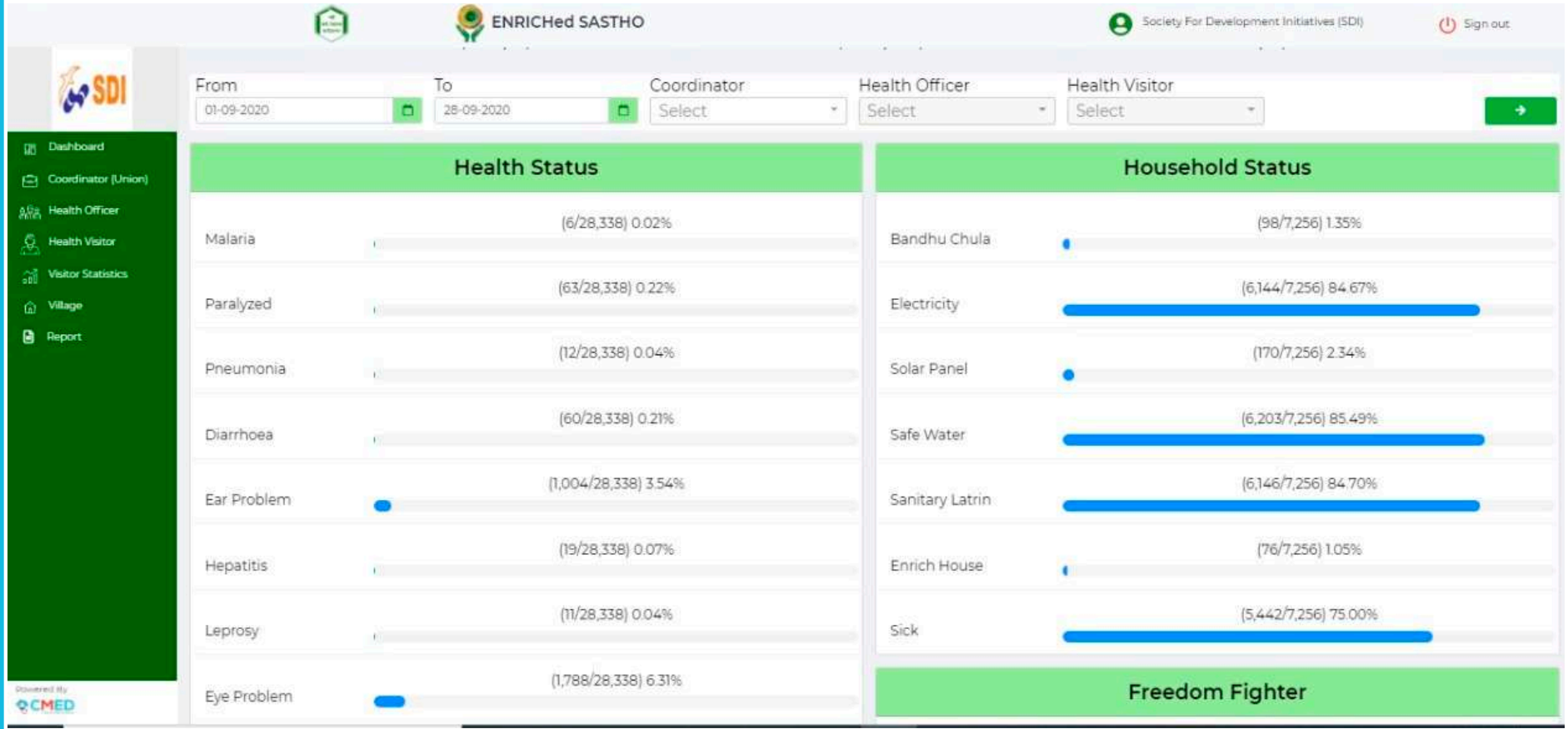
স্বাক্ষর
Dr. yasin Arafat

CMED Platform for PKSF (Monitoring & Management Dashboard)



CMED Digital Healthcare Platform for Primary and Preventive Care

CMED Platform for PKSF (Monitoring & Management Dashboard)



CMED Platform for PKSF (Monitoring & Management Dashboard)



CMED Digital Healthcare Service Delivery














উদ্বোধনী অনুষ্ঠান
CMED

ডিজিটাল স্বাস্থ্যসেবা (জিপি মডেল)



ঠিকানাঃ
আলহাজ্ব এম এ রশীদ মা, শিশু ও চক্ষু হাসপাতাল
মালঞ্চ, মেলান্দহ উপজেলা, জামালপুর।

জনগণের দোরগোড়ায় ডিজিটাল স্বাস্থ্যসেবা

জিপি সেন্টারের ফোন নম্বরঃ ০১৯০৭১৩০১৭০
www.cmed.com.bd

ডিজিটাল স্বাস্থ্যসেবা (জিপি মডেল)





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www.cmed.com.bd

ডিজিটাল স্বাস্থ্যসেবা (জিপি মডেল)







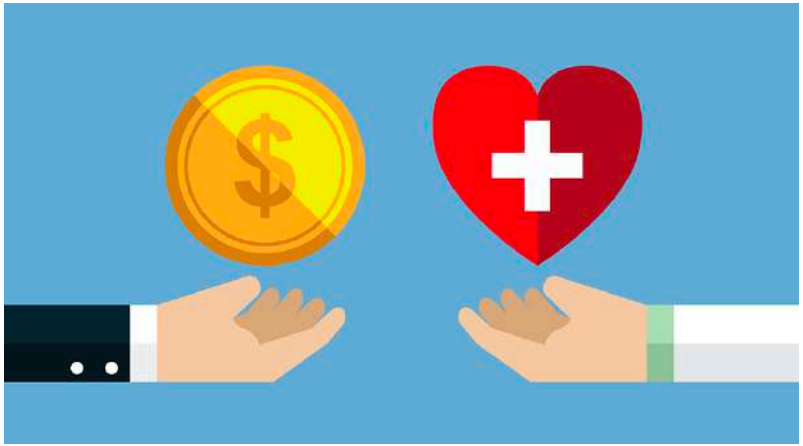


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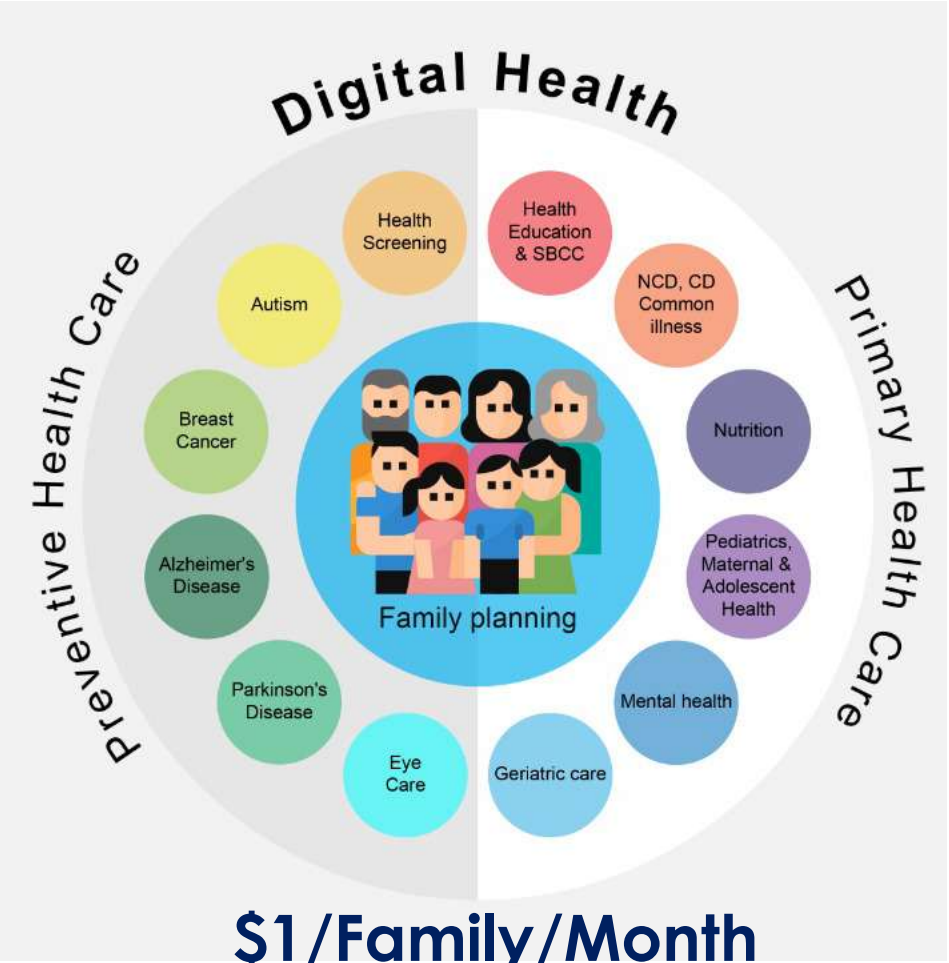
জিপি সেন্টারের ফোন নম্বরঃ ০১৯০৭১৩০১৭০
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General Practitioner (GP) Model using CMED Digital Healthcare Platform



Cost of PHC





$1/Family/Month

Scaling of CMED Digital Health Service



SDG





Achievement & Impact

1.3 Million+
Monthly Users by 2020

1,000+
Jobs Created

1100+
Villages Covered (51 Unions)

2 Million+ Users served during COVID-19 through 10K+ Govt. Health Workers

Award & Recognition

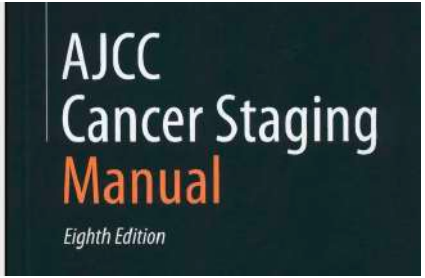
BANGLADESH INNOVATION AWARD 2019
Winner Healthcare

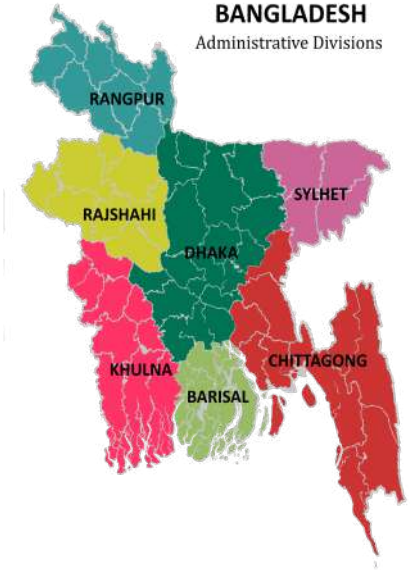
APICTA Asia Pacific ICT Alliance AWARDS
Merit Winner 2019

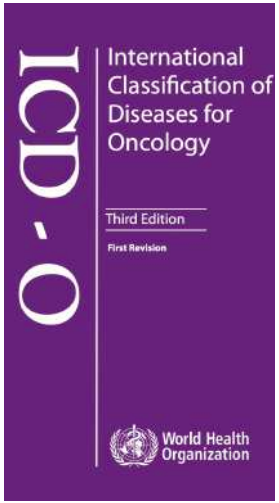
DBS · NUS Social Venture Challenge Asia
Winner: THF Innovation Award 2018

seedstarsworld*
Global Innovation Winner 2018

Central Cancer Registry (CCR) and Management System







Activates in 2019-2020

- ✓ Research on Cancer Registry System
- ✓ 5 workshops with Oncology Communities for requirement analysis on Cancer Registry
- ✓ Design Cancer Registry System for Bangladesh
- ✓ Develop clinical diagnosis mapping algorithm for all (84) types of cancer for CCR according to AJCC Cancer Staging Manual and ICD O
- ✓ Developed CCR version 1 and had 10 workshops with Oncology Communities on it
- ✓ Launched December 7, 2019 (Version 1) for Dhaka
- ✓ CCR is running in 46 Doctor Chambers/hospitals

Central Cancer Registry (CCR) and Management System

৩০-০১-২০২১ সংবাদ অনুষ্ঠান বুলেটিন এক্সক্লুসিভ সরাসরি সম্প্রচার যোগাযোগ অনুসন্ধান

সময়

Dec. 2019

সময় সংবাদ

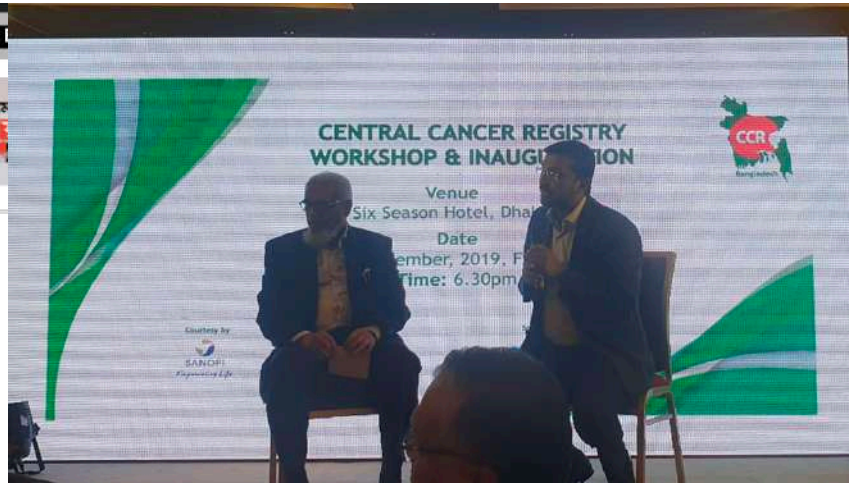
আপডেট
০৭-১২-২০১৯, ১১:৪৪

ক্যান্সার চিকিৎসায় বছরে সরকারের ব্যয় ১৫ কোটি টাকা



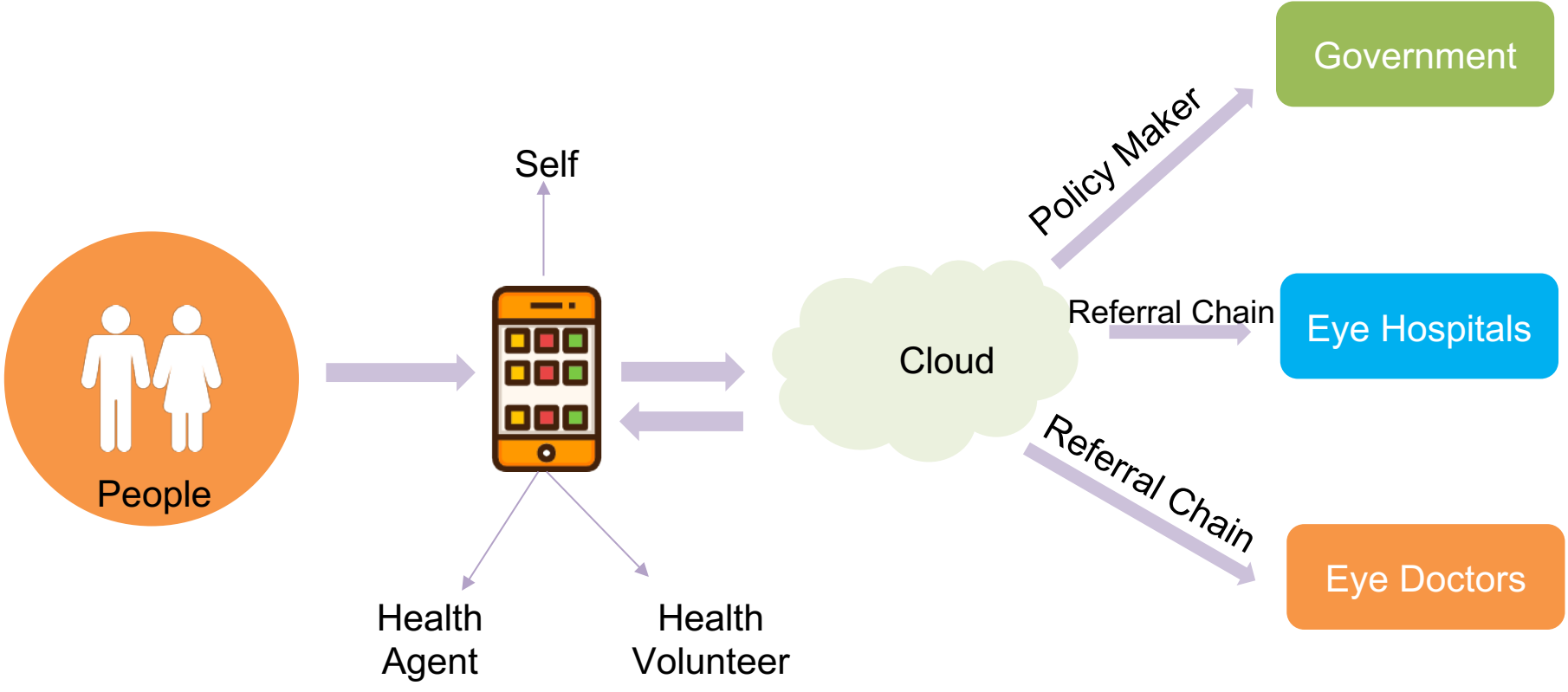
ক্যান্সারে আক্রান্ত শুধু এক হাজার রোগীর জন্য ওষুধ বাবদই প্রতিবছর সরকারের ব্যয় হচ্ছে ১৫ কোটি টাকা। বিশ্বব্যাপী মরণব্যাধি ক্যান্সারে আক্রান্ত মানুষের সংখ্যা প্রতিনিয়ত বৃদ্ধি পেলেও বাংলাদেশের এ বিষয়ে নিজস্ব কোনো পরিসংখ্যান নেই। এমনটিই জানিয়েছেন জাতীয় ক্যান্সার হাসপাতাল ও

আরও পড়ুন
যে ৩ অভ্যাসে ক্যান্সার থেকে মিলবে চিরমুক্তি
ব্রেস্ট ক্যান্সার বেশি হয় যে নারীদের
মেকআপে হতে পারে ক্যান্সার





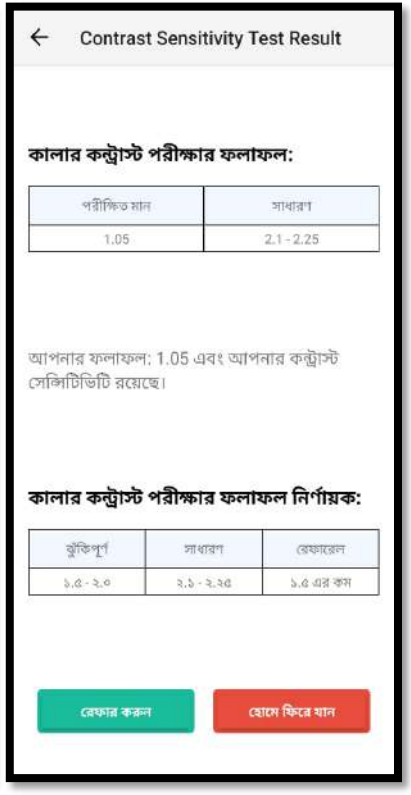
Virtual Eye Doc: A Cloud Based Smart Application for Eye Care in Bangladesh



GET IT ON Google Play

Virtual Eye Doc: A Cloud Based Smart Application for Eye Care in Bangladesh









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ICT DIVISION
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GET IT ON Google Play

PVDoc: Parkinson's Virtual Doctor

Cloud based AI Driven Screening and Monitoring System for Parkinson's Diseases

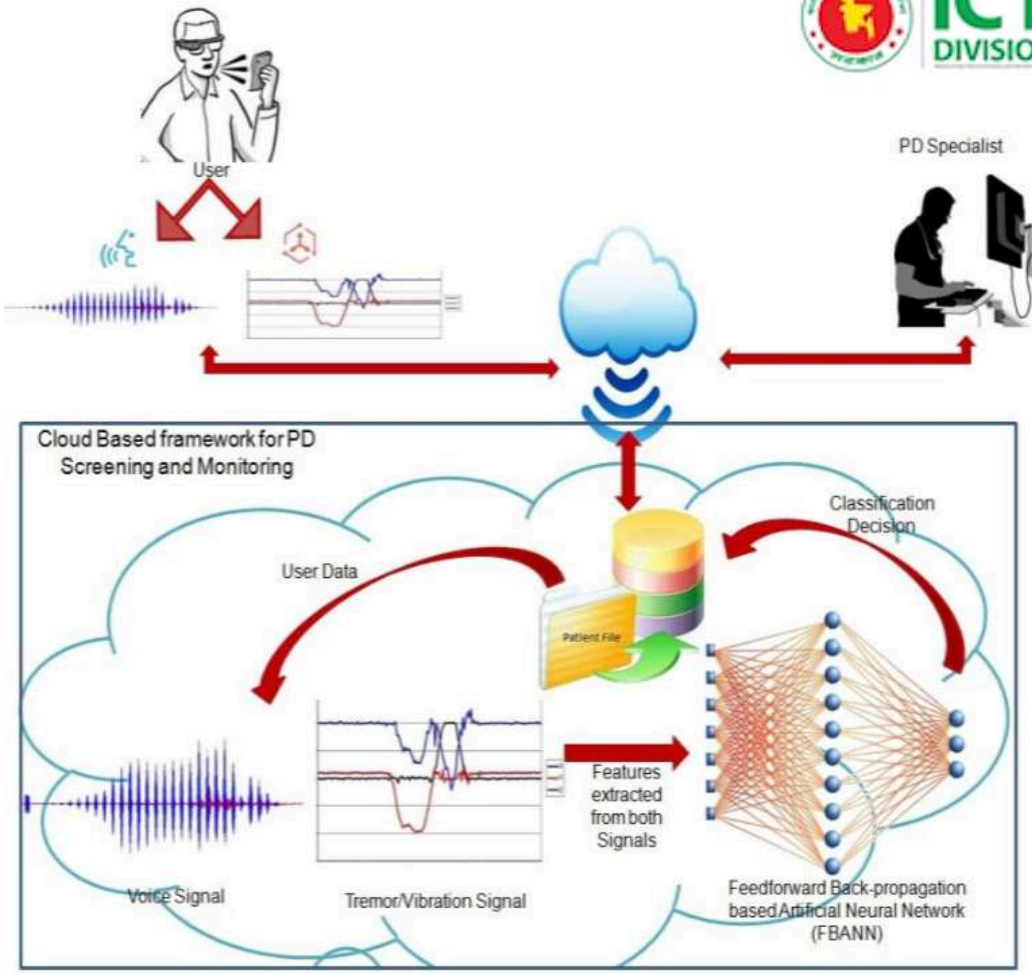
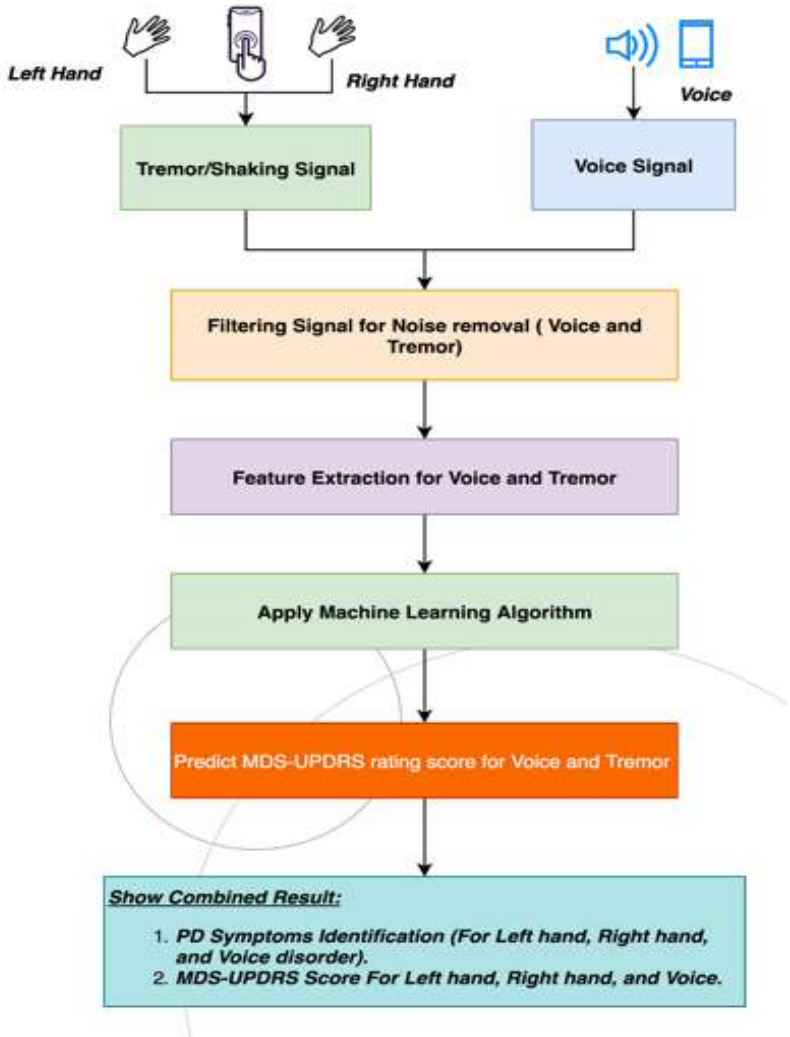


Figure 1. PVDoctor framework with its primary users.

PVDoc: Parkinson's Virtual Doctor



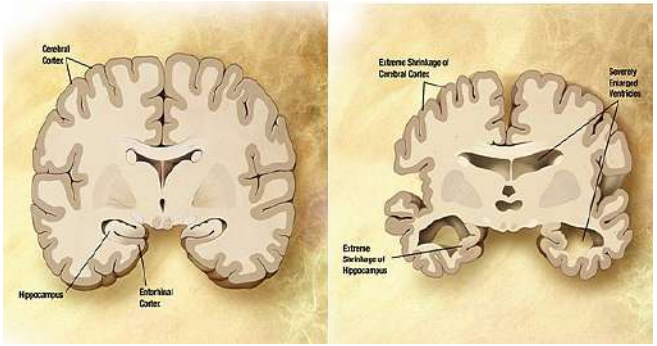
Journal

- *Md. Sakibur Rahman Sajal, Md. Tanvir Ehsan, Ravi Vaidyanathan, Shouyan Wang, Tipu Z. Aziz, Khondaker Abdullah Al Mamun, "**Telemonitoring Parkinson's disease using machine learning by combining tremor and voice analysis**". Brain Informatics 7(1): 1-11 (2020).*

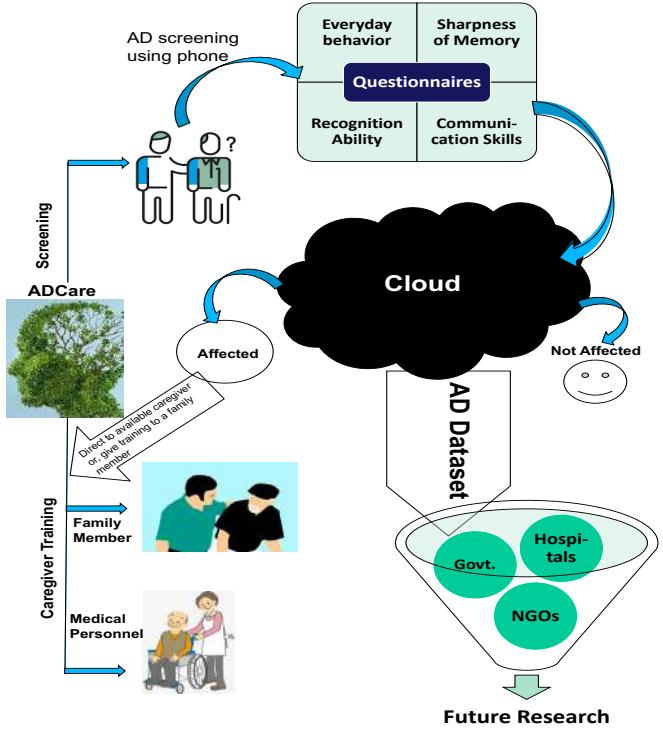
Conferences

- *Md. Tanvir Ehsan, Shehan Irteza Pranto, Khondaker A. Mamun,"**Real-Time Screening of Parkinson's Disease based on Speech Analysis using Smartphone**" 10th International IEEE/EMBS Conference on Neural Engineering (NER), 4-6 May 2021, Virtual. (Submitted)*
- *Md Tanvir Ehsan, Md Sakibur Rahman Sajal, and Khondaker A. Mamun. "**An automated cloud-based tool for Screening of Parkinson's disease in Bangladesh**." 2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST). IEEE, 2021.*
- *Md. Sakibur Rahman Sajal, Md. Tanvir Ehsan, Ravi Vaidyanathan, Shouyan Wang, Tipu Z. Aziz, Khondaker Abdullah Al Mamun, "**UPDRS Label Assignment by Analyzing Accelerometer Sensor Data Collected from Conventional Smartphones**. Brain Informatics 2020: 173-182, on 15 September 2020.*

ADCare: Smart tools for screening, resource mapping, and increasing awareness of Alzheimer's Disease among population in Bangladesh







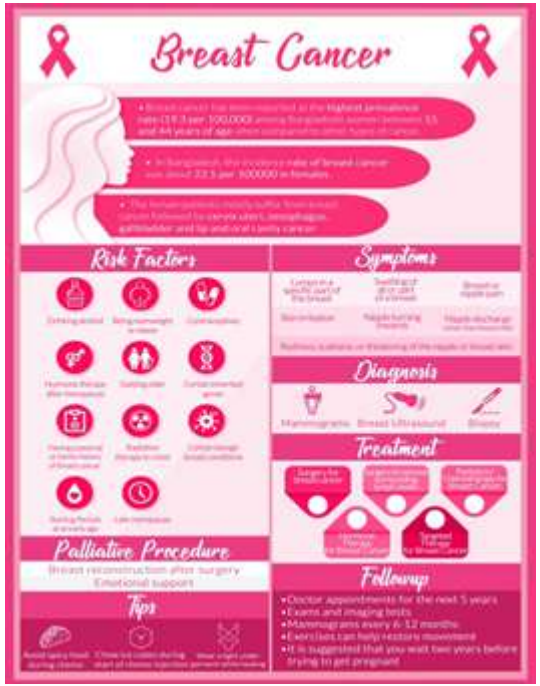
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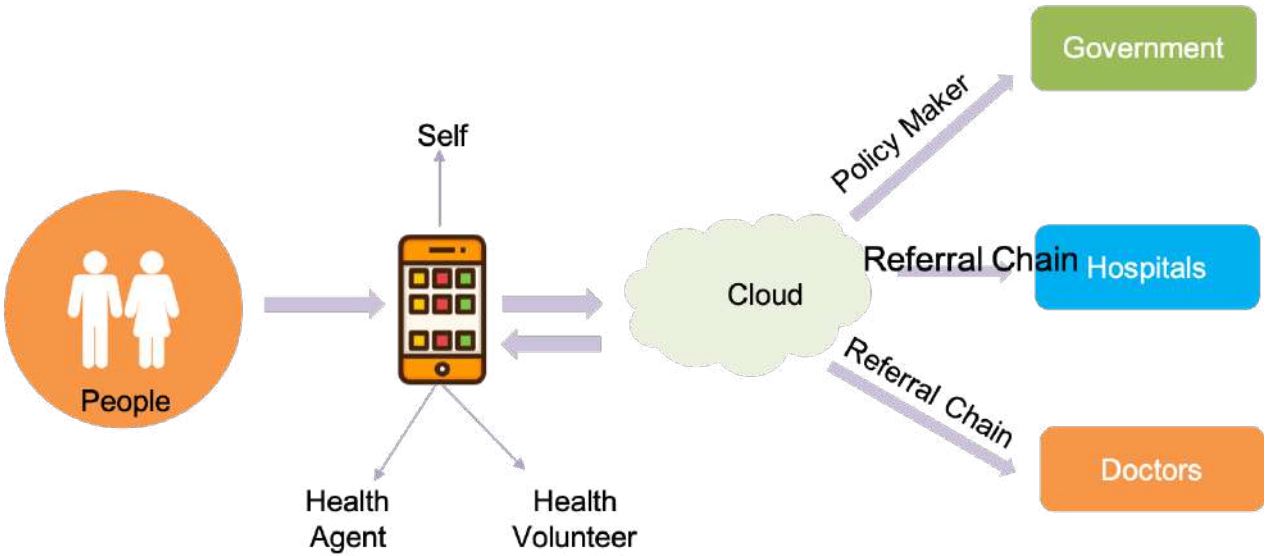
United International University

ICT DIVISION
FUTURE IS HERE

Can-Aware

An interactive system for cancer awareness, education, screening and referral in Bangladesh





Assistive Technologies



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Advanced Intelligent Multidisciplinary Systems Lab

Bolte Chai Plus

An intelligent integrated cloud based platform to augment communication and speech for verbally challenged Autism and NDD children

Tauri Foundation
Committed to Secure Child Rights & Inclusive Society

CMED
Connecting People, Saving Lives

UIU
United International University
QUEST FOR EXCELLENCE

grameenphone

Holland Bloorview
Kids Rehabilitation Hospital

prism lab

UNIVERSITY OF TORONTO

নিউরো-ডেভেলপমেন্টাল প্রতিবন্ধী সুরক্ষা ট্রাস্ট

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Bolte Chai

Ensures my RIGHT to express my DESIRE...

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Problem Statement



1%

of the world population are suffering from Autism Spectrum Disorder among which,

25% are Non-verbal

Nearly

Half a Million

children are Non-verbal in **Bangladesh**



Difficulties in communication results,



ANGER

irritating behavior & stubbornness

ANXIETY

loneliness & distress



Bolte Chai

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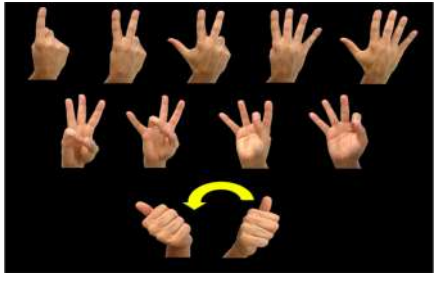
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Existing Solutions

Picture Exchange Communication System (PECS).



Sign Language





Augmentative and Alternative Communication (AAC)

Mobile application based AAC systems in Developed Countries

avaz

TouchChat

Voice4u

Bolte Chai: Digital Communication Tool for Enhancing Communication for Autistic and NDD Non-Verbal Children



Bolte Chai Custom device

- Not fully customizable
- No collaborative platform
- No proper monitoring and report generation facility
- Not suitable for local context and language of Bangladesh

&

Costly

Bolte Chai

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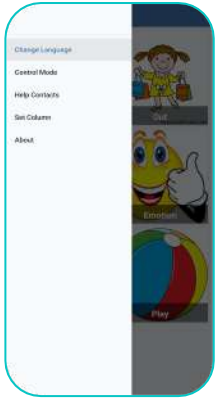
Bolte Chai

Desire
Need
Emergency



GET IT ON Google Play















Using Bolte Chai nonverbal children can express their desires and it improves communications.

Bolte Chai already proved as a tool for speech enhancement and *two nonverbal child are improved to verbal stage using it.*

Bolte Chai Journey Demo and Success Story : https://www.youtube.com/watch?v=kD6HEnxY_eQ



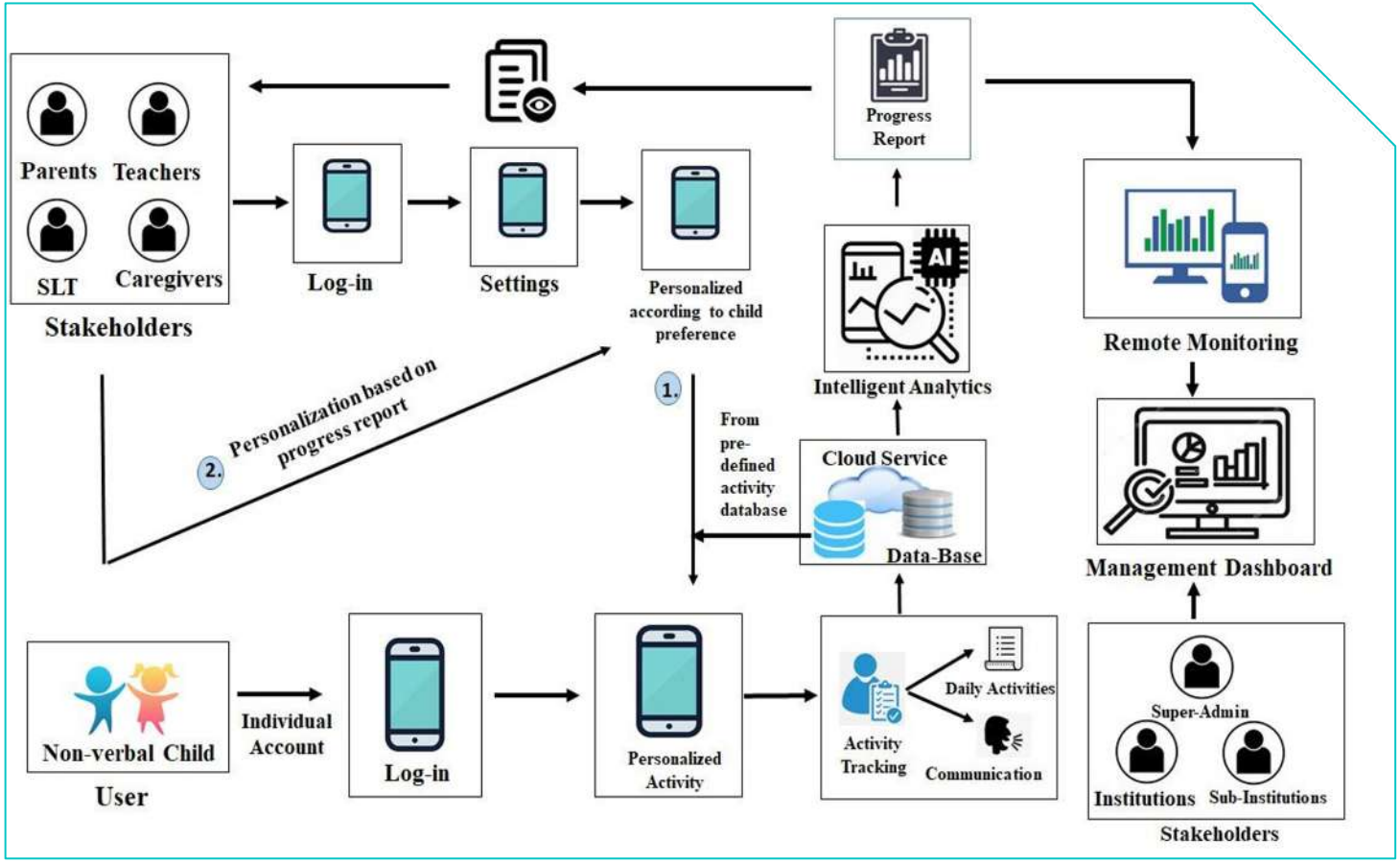
Bolte Chai

Ensures my RIGHT to express my DESIRE...

Bolte Chai Plus

The Updated Version of **Bolte Chai**

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Features

- ❑ Individual user account
- ❑ Personalization of activities
- ❑ Pre-defined Activity Database in Cloud
- ❑ App & Integrated Website
- ❑ Activities Tracking
- ❑ Intelligent Analytics
- ❑ Progress Monitoring & Report
- ❑ User Management & Dashboard
- ❑ Collaboration among stakeholders

Bolte Chai Plus-Proposed Solution : https://www.youtube.com/watch?v=nUlDr0cBloQ&feature=youtu.be

Bolte Chai

Ensures my RIGHT to express my DESIRE...

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Research & Collaboration

Publications

1. Khan, Md Nasfikur R., et al. "Bolte Chai: An augmentative and alternative communication device for enhancing communication for nonverbal children." *2016 International Conference on Medical Engineering, Health Informatics and Technology (MediTec)*. IEEE, 2016.

2. Khan, Md Nasfikur R., et al. "'Bolte Chai'—An Android application for verbally challenged children." *2017 4th International Conference on Advances in Electrical Engineering (ICAEE)*. IEEE, 2017.

3. Farzana, Walia, et al. "Communication Support Utilizing AAC for Verbally Challenged Children in Developing Countries During COVID-19 Pandemic." *International Conference on Human-Computer Interaction*. Springer, Cham, 2020.

4. Farzana, Walia, et al. "An Evaluation of Augmentative and Alternative Communication Research for ASD Children in Developing Countries: Benefits and Barriers." *International Conference on Human-Computer Interaction*. Springer, Cham, 2020.

5. Farzana, Walia, et al. "Technological Evolvement in AAC Modalities to Foster Communications of Verbally Challenged ASD Children: A Systematic Review." IEEE Access (2021).

Partners

CMED
Connecting People, Saving Lives

Tauri Foundation
Committed to Secure Child Rights & Inclusive Society

prism lab

Holland Bloorview
Kids Rehabilitation Hospital

UNIVERSITY OF TORONTO

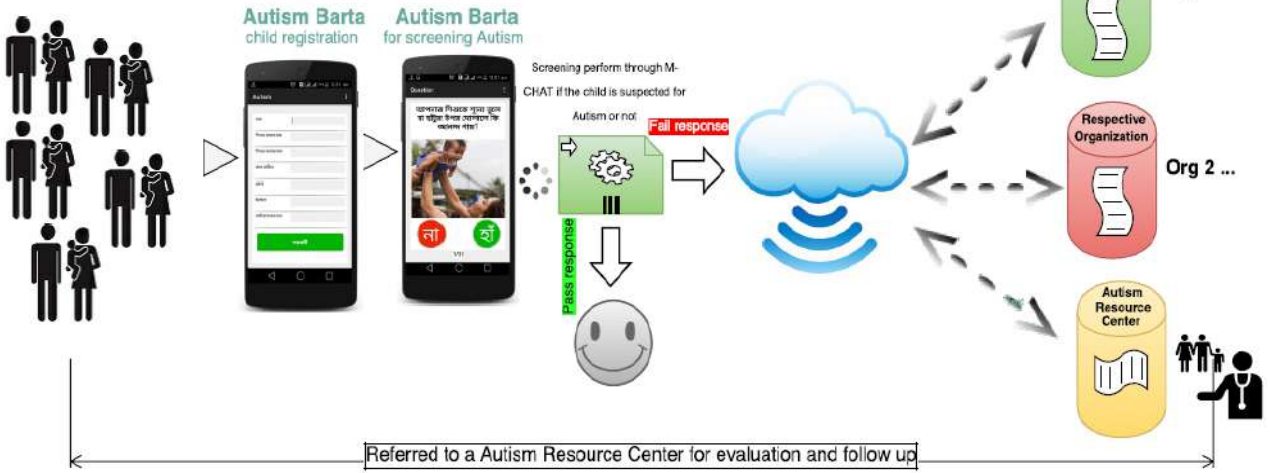
নিউরো-ডেভেলপমেন্টাল প্রতিবন্ধী সুরক্ষা ট্রাস্ট

জাতীয় প্রতিবন্ধী উন্নয়ন ফাউন্ডেশন
সমাজকল্যাণ মন্ত্রণালয়

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Autism Barta: Automated Autism Screening Tools





INSTITUTE FOR PAEDIATRIC NEUROSURGERY & AUTISM (IPNA)
BANGABANDHU SHEIKH MUJIB MEDICAL UNIVERSITY (BSMMU)

BCI APPLICATIONS



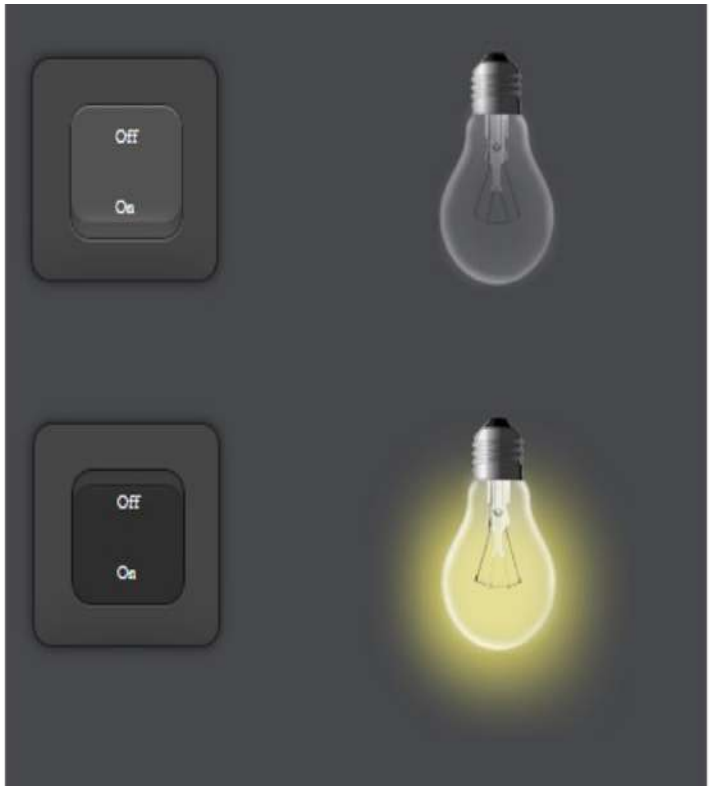
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NeuroLite : Brain Computer Interface Driven Applications

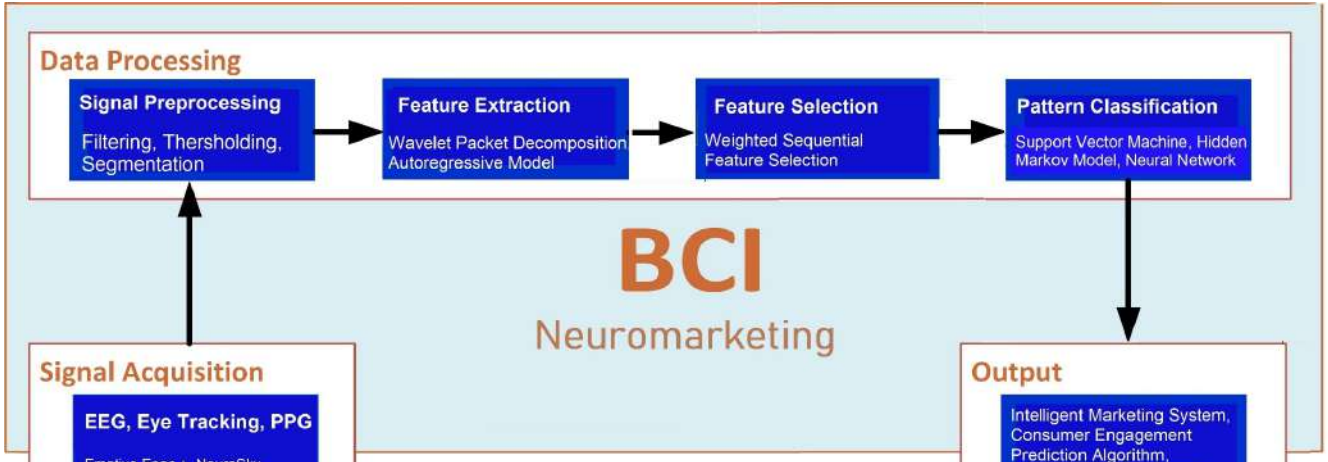








Development of a BCI based Neuromarketing System





Brain Signal (EEG)







Physiological Signal



Eye Tracker

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Imperial College London

UNIVERSITY OF TORONTO

Development of a BCI based Neuromarketing System

Rawnaque *et al. Brain Inf.* (2020) 7:10
https://doi.org/10.1186/s40708-020-00109-x

Brain Informatics

REVIEW Open Access

Technological advancements and opportunities in Neuromarketing: a systematic review

Ferdousi Sabera Rawnaque[1*], Khandoker Mahmudur Rahman[2], Syed Ferhat Anwar[3], Ravi Vaidyanathan[4], Tom Chau[5], Farhana Sarker[6] and Khondaker Abdullah Al Mamun[1,7]

Abstract

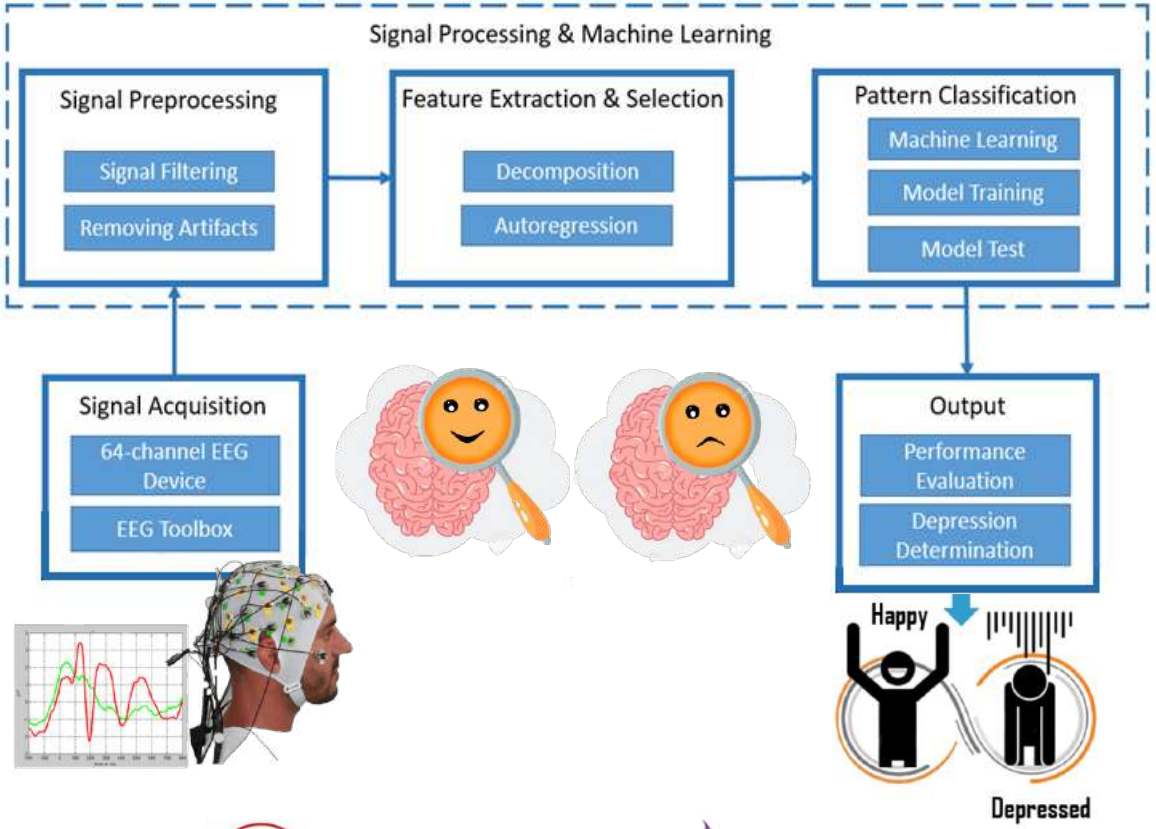
Neuromarketing has become an academic and commercial area of interest, as the advancements in neural recording techniques and interpreting algorithms have made it an effective tool for recognizing the unspoken response of consumers to the marketing stimuli. This article presents the very first systematic review of the technological advancements in Neuromarketing field over the last 5 years. For this purpose, authors have selected and reviewed a

F. R. Mashrur, M. T. I. Miya, F. S. Rawnaque, R. Vaidyanathan, S. F. Anwar, F. Sarker, and K. A. Mamun "MarketBrain: An EEG based intelligent consumer preference prediction system," in 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2021), (Submitted).

DEEP-DEPRESSION: Design and Development of Brain Computer Interface (BCI) Based System for Identifying Depression Biomarker Using Neural Connectivity Analysis

DEEP DEPRESSION
STAY STRONG, STAY HAPPY

Drafted a systematic review on the BCI based depression biomarkers identification and its applications



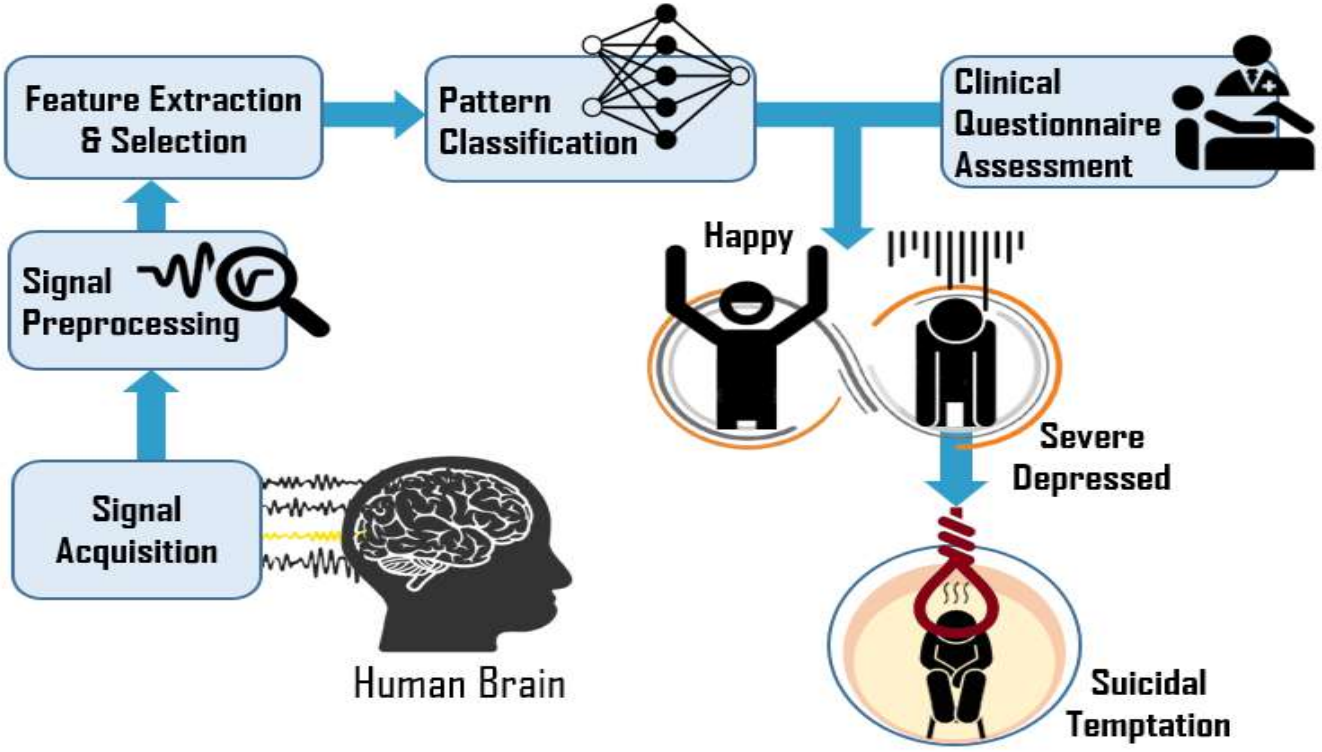
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IUB
Independent University, Bangladesh

Imperial College London

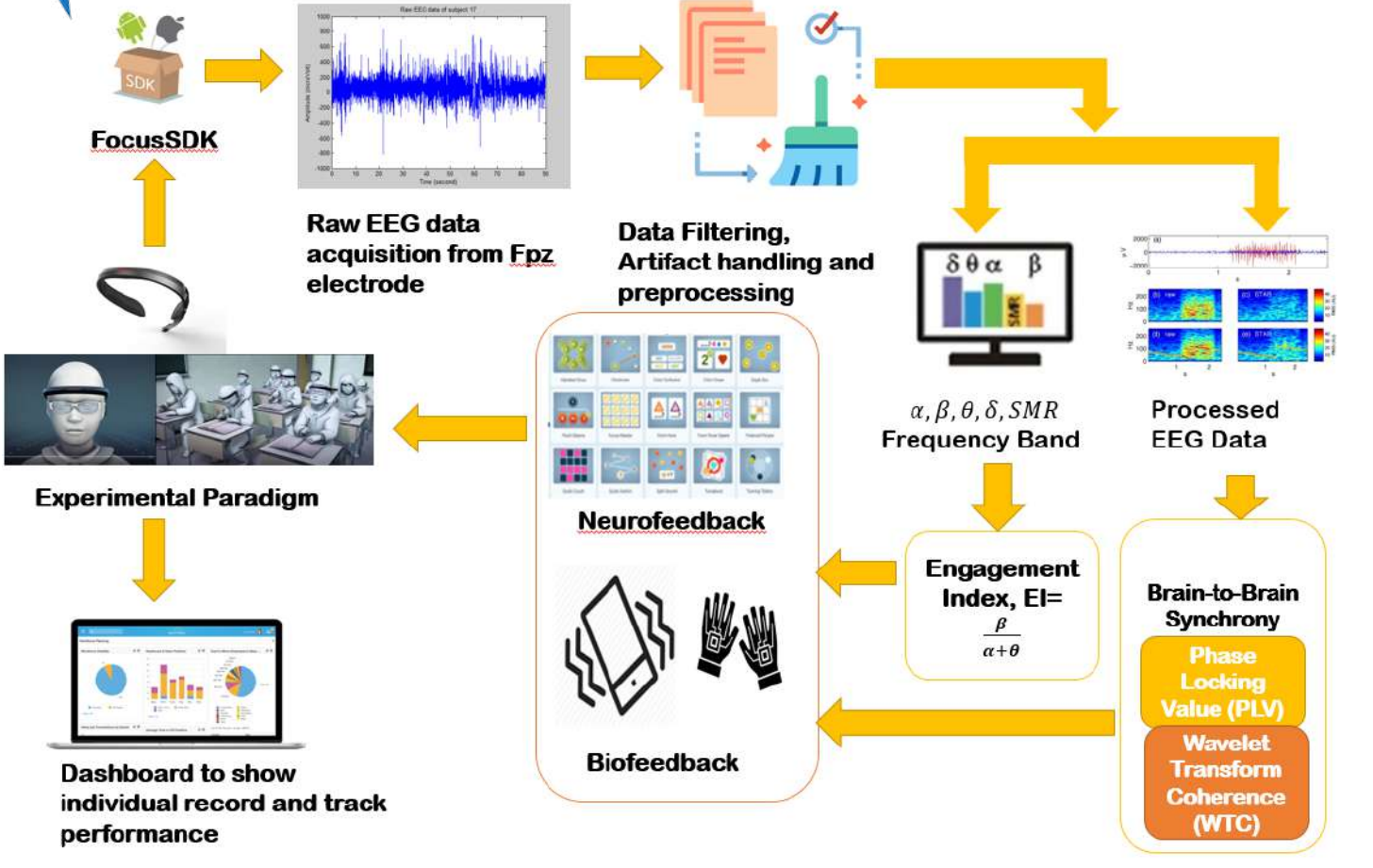
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Design and Development of Brain Computer Interface (BCI) Based System for Identifying Suicidal Biomarkers using Neural Connectivity Analysis



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United International University
National Institute of Mental Health & Hospital, NIMHH
Medical College for Women & Hospital
IUB Independent University, Bangladesh
Imperial College London
University of Toronto

FocusMind: An Integrated BCI and Neurofeedback-based Learning Enhancement System Utilizing Brain-to-Brain Synchrony



BrainCo
YOUR BRAIN CONTROLS EVERYTHING

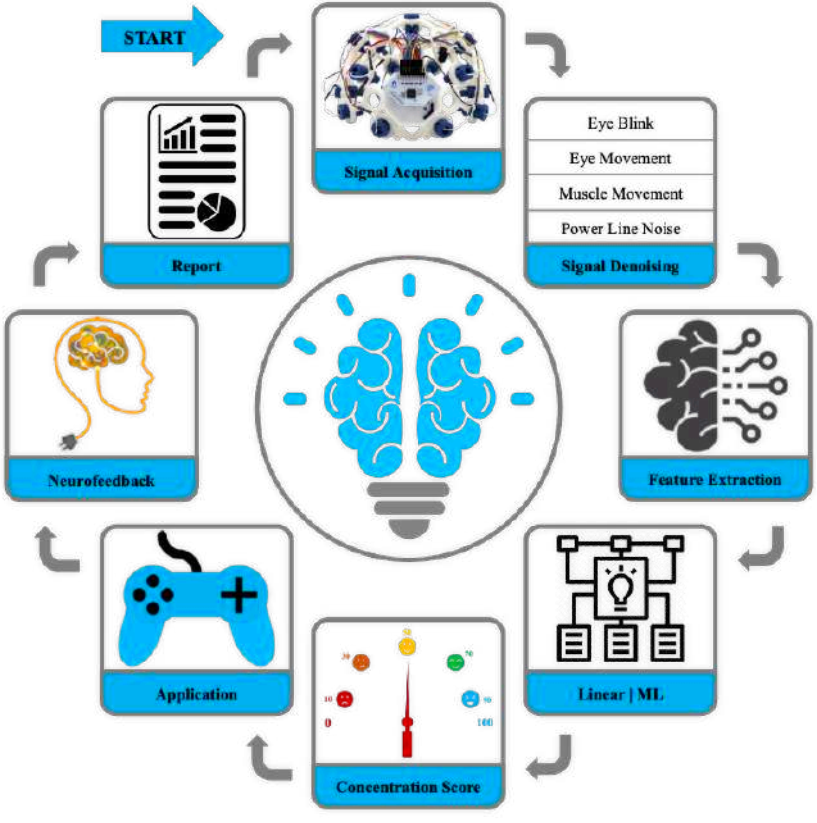
United International University
QUEST FOR EXCELLENCE

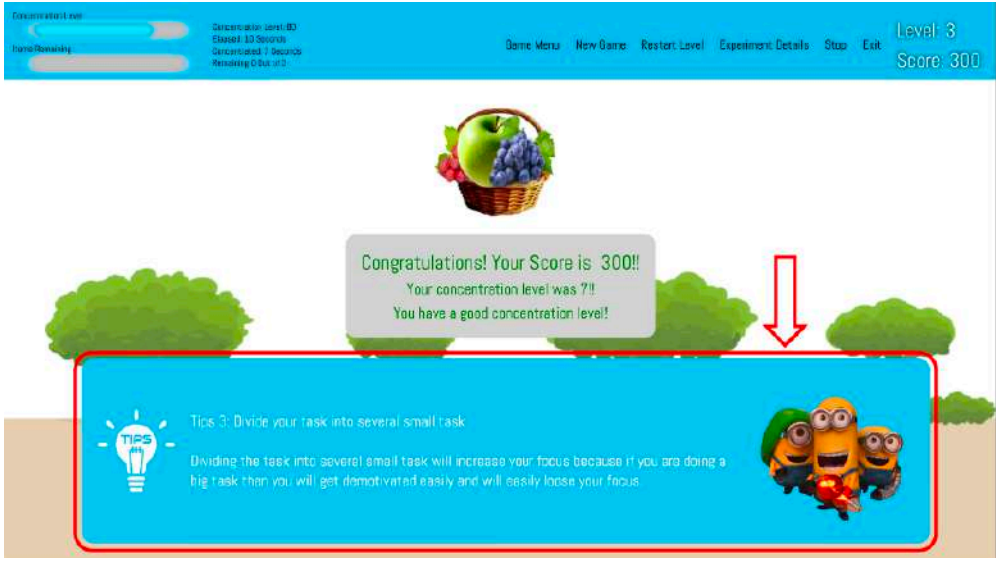
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ICT DIVISION
FUTURE IS HERE

Tahnia Nazneen, Iffath Binta Islam, Wasifa Jamal, Ravi Vaidyanathan, Tom Chau, Khondaker A. Mamun, "Neural recording based brain-to-brain synchrony analysis on multidisciplinary human interactions for understanding brain dynamics: A systematic review" NeuroImage (under review)

Concentrate.Today – A Real-Time Neurofeedback Based Brain-Computer Interface (BCI) Enabled System for Enhancing Concentration Level



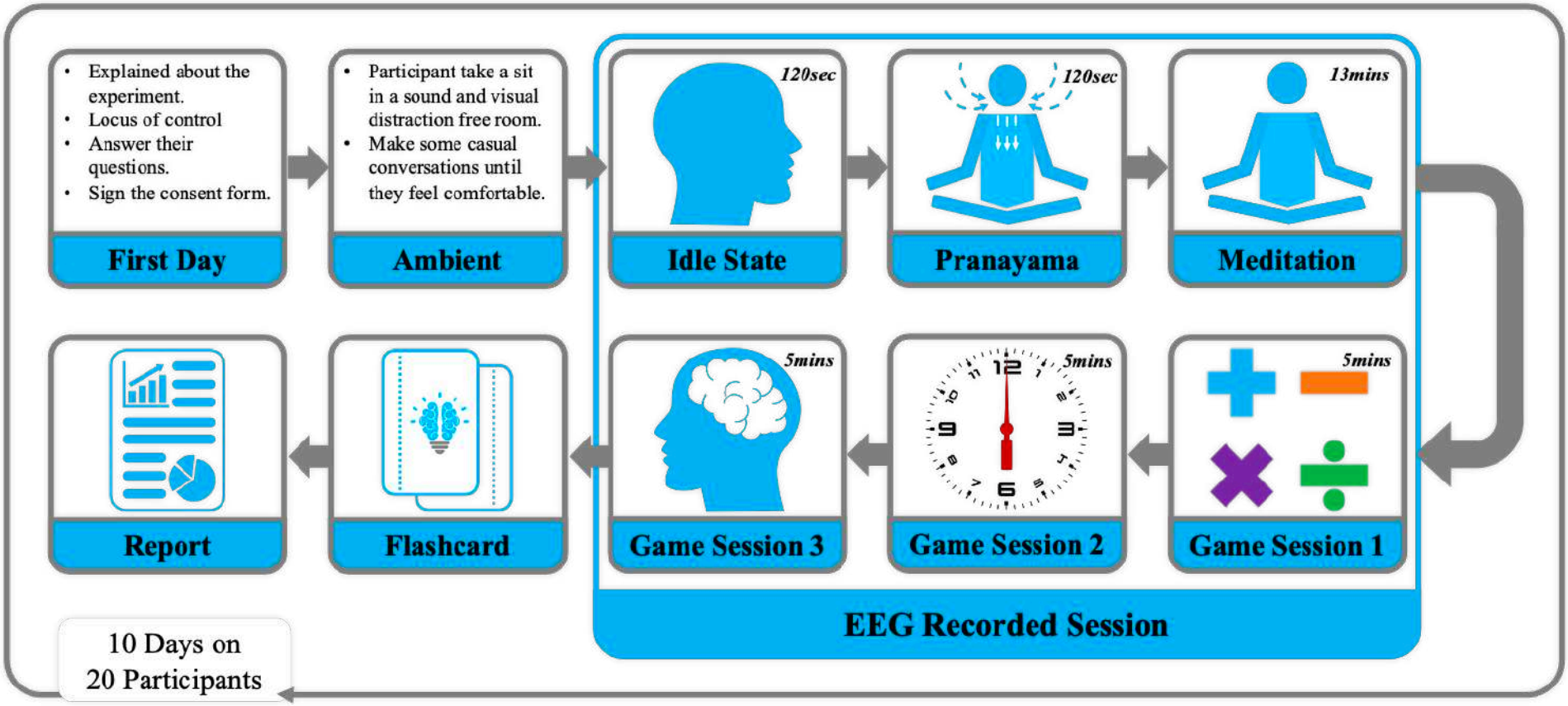


OpenBCI

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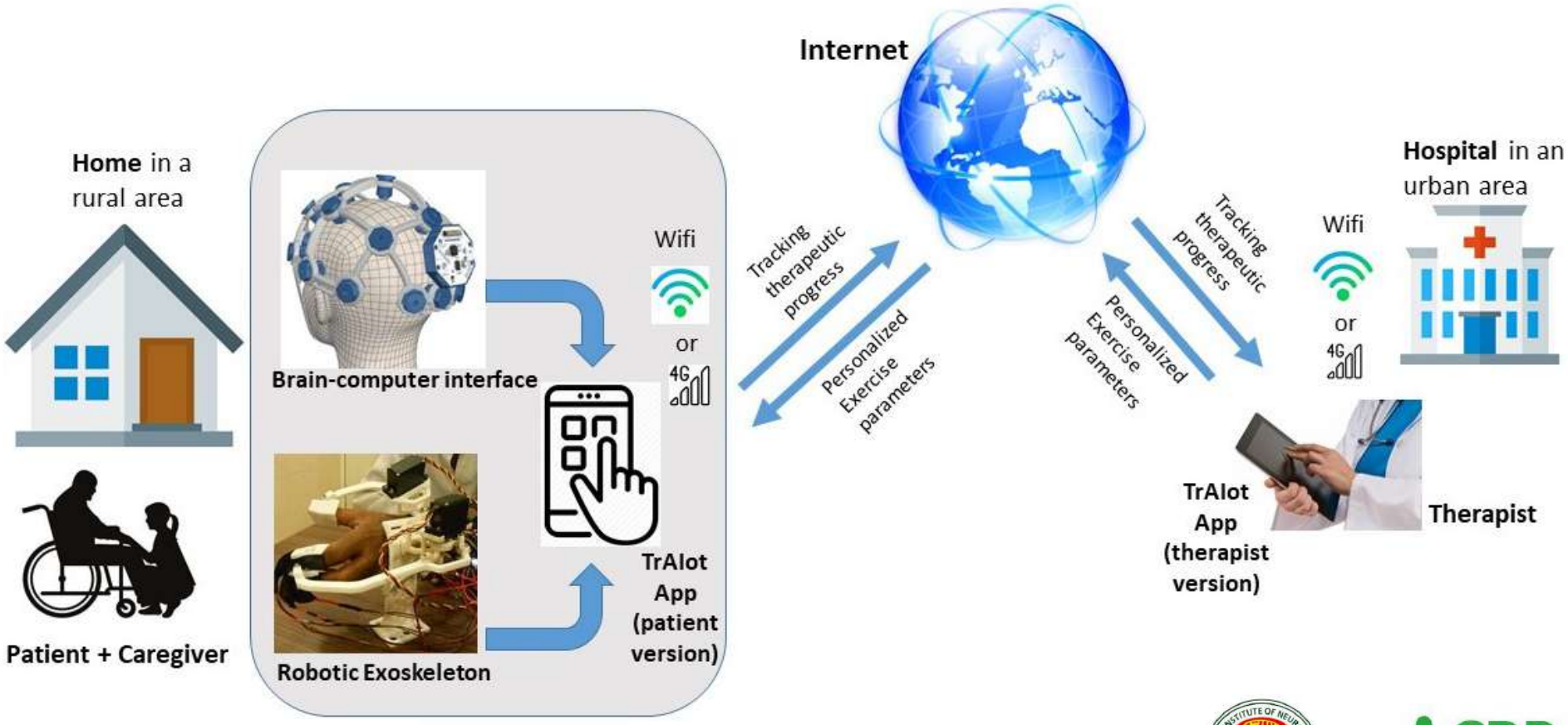
ICT DIVISION
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Concentrate.Today – A Real-Time Neurofeedback Based Brain-Computer Interface (BCI) Enabled System for Enhancing Concentration Level



Rana Md Shahariar Parbez and Khondaker A. Mamun, "BlinkFruity: A Real-Time EEG Based Neurofeedback Game for Brain-Computer Interface", 2nd International Conference on Advanced Information and Communication Technology 2020 (ICAICT 2020), 28-29 November 2020. **Also a Game Developed**

BRIGHT: Brain-Robot Interface Guided IoT for Home-based Post-stroke Telerehabilitation



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Advanced Intelligent Multidisciplinary Systems Lab

University of Essex

Ulster University

CRP
Centre for the Rehabilitation of the Paralysed

ROBOTIC APPLICATIONS

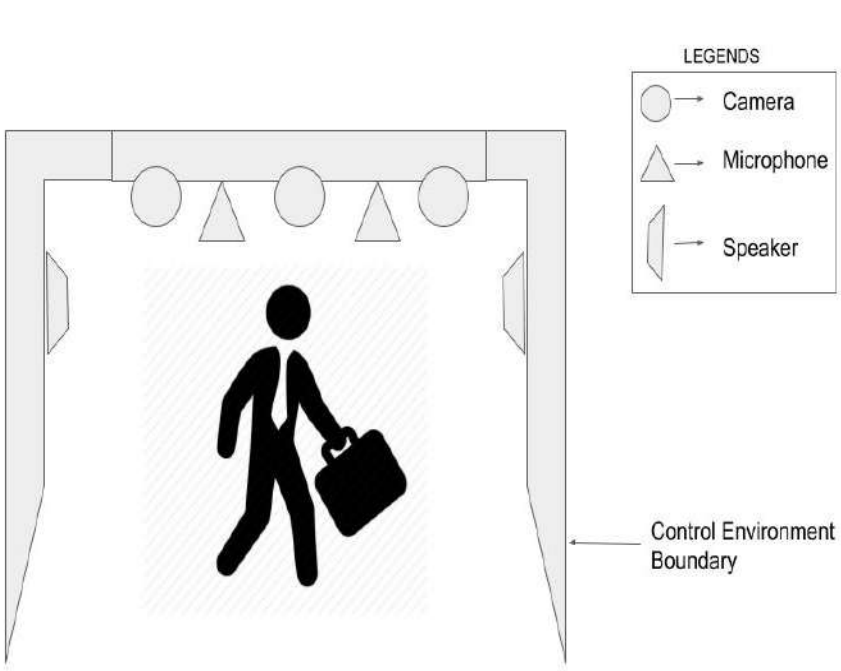


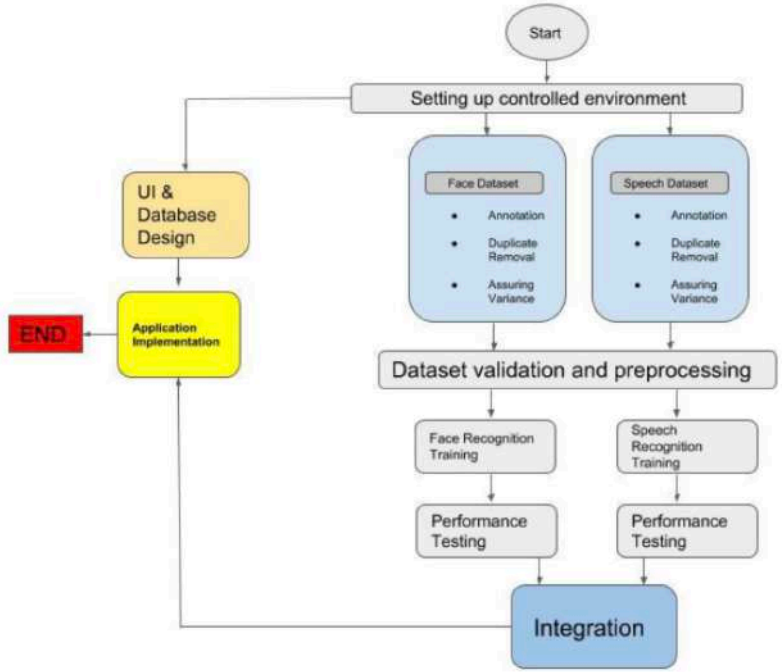
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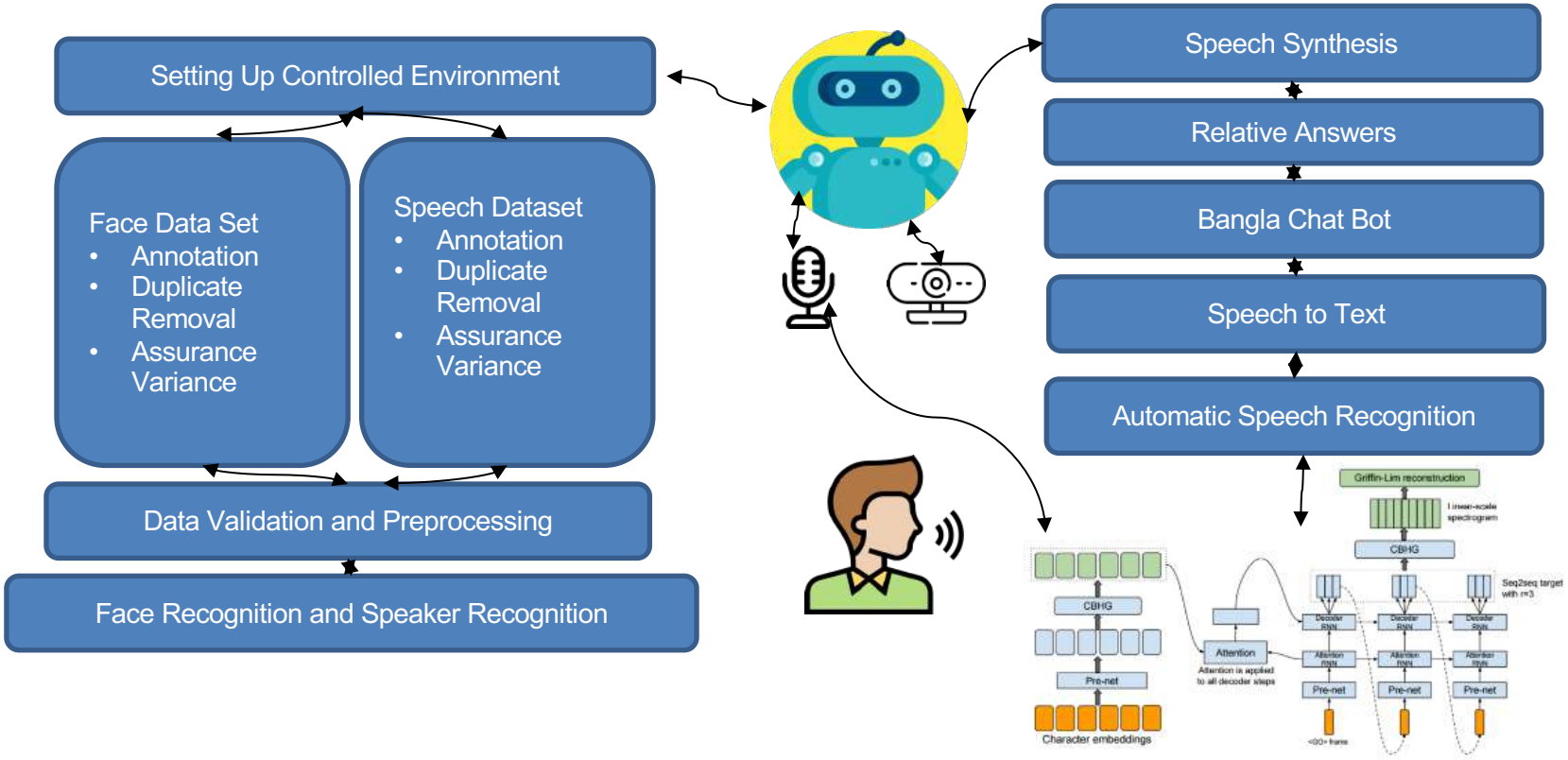
Smart Reception

An artificial intelligence driven receptionist to automate reception services using speech, speaker and face recognition





Smart Reception



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UIU
UNITED INTERNATIONAL UNIVERSITY

ICT DIVISION
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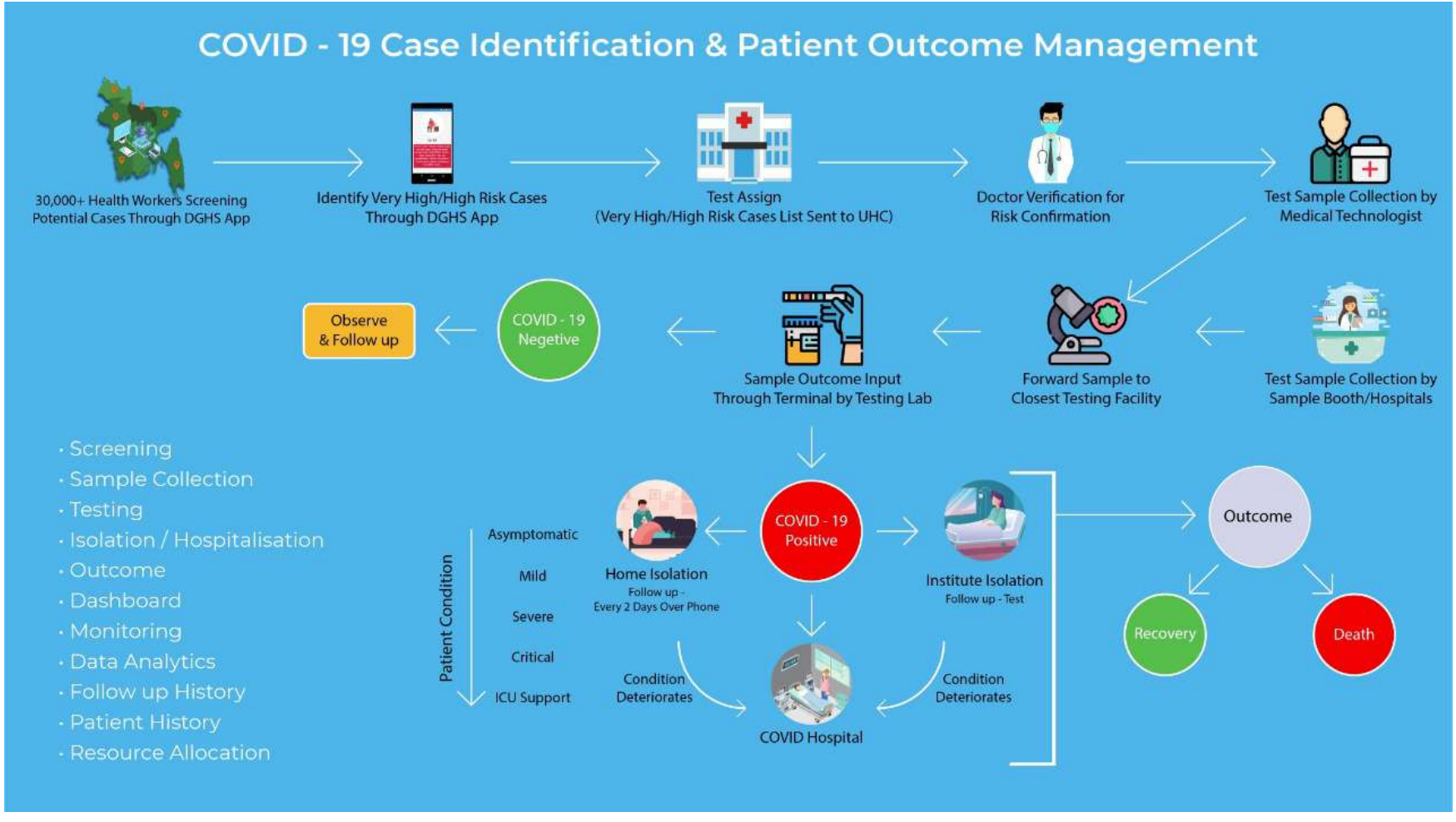
COVID 19 APPLICATIONS



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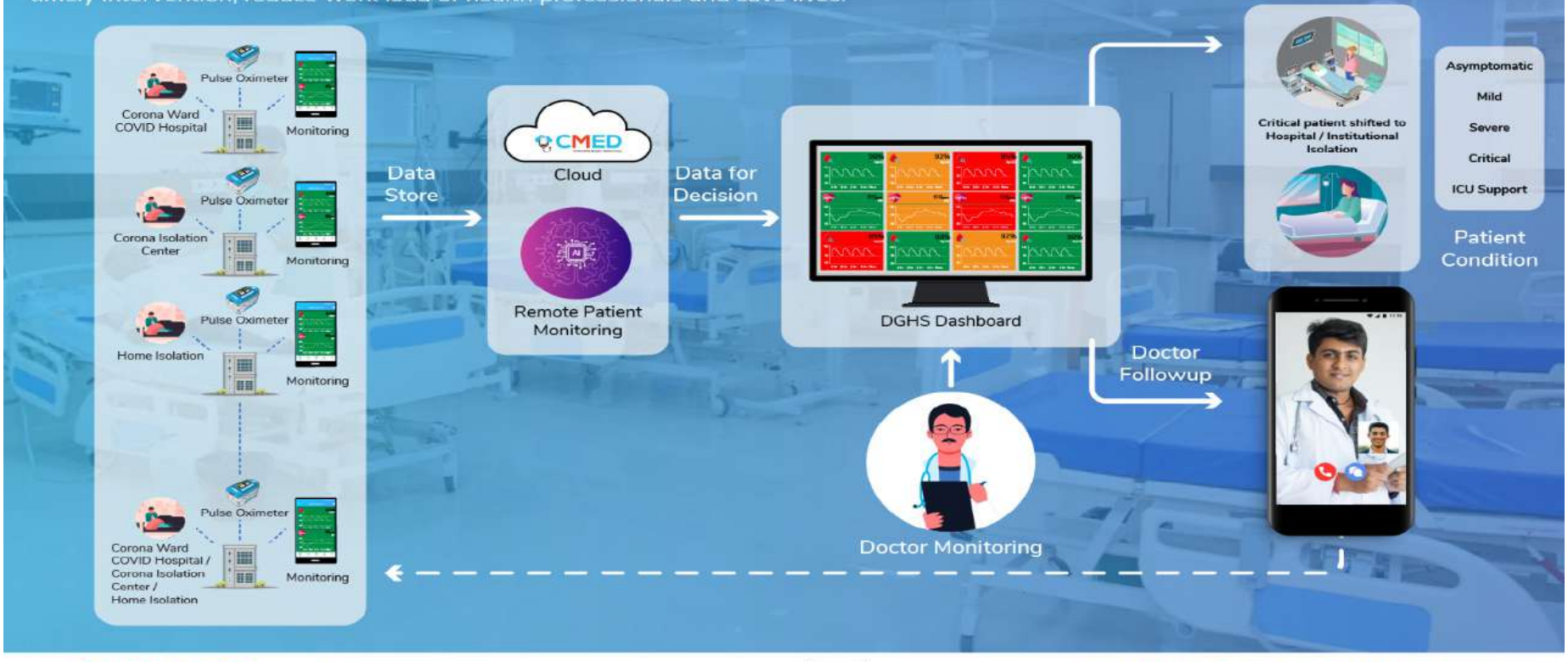
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AI Based COVID-19 Digital Surveillance Platform



COVID-19 Digital Remote Patient Monitoring & Virtual ICU System

Digital remote patient monitoring and virtual ICU system: The system uses technology to monitor patient health outside of a traditional clinical settings, such as in home or in a hospital. It collects level of oxygen in patient's blood along with heart rate, alerts the health care provider if oxygen levels drop below safe levels and assess the need for clinical intervention such as supplemental oxygen. This system can play significant role for monitoring covid-19 patients in different settings like, COVID-19 hospital, isolation center and home isolation to make timely intervention, reduce work load of health professionals and save lives.



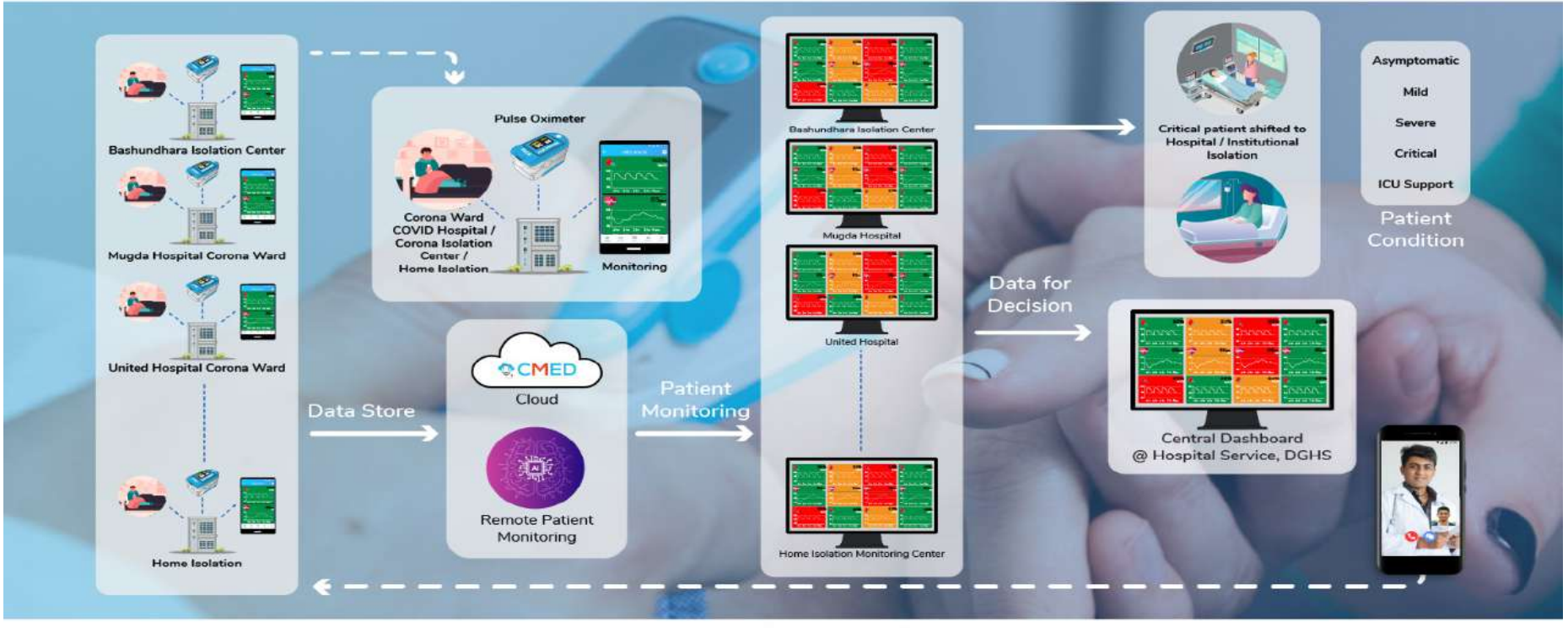
CMED
Connecting People, Saving Lives

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CE FDA

Apt # C-5 & C-4, House # 761, Safura Green,
Satmasjid Road, Dhanmondi, Dhaka-1209

COVID-19 Digital Remote Patient Monitoring & Virtual ICU System



Patient Benefits:

- Easy to use
- Fast, accurate, continuous monitoring
- Early detection and earlier intervention
- Minimized complications
- Reduction in hospitalization
- Improved patient management

Doctors and Hospital benefits:

- Continuous patient monitoring data streaming could provide real-time patient supervision.
- Balances out the workload and time; no need for rounds to assess patients in such cases.
- Better utilization of medical resources such as the equipment, medical staff, and others.

Regulatory Benefits:

- Central monitoring of covid-19 patients
- Optimize resource mobilization
- Data based decision and intervention
- Better access to healthcare system

CMED
Connecting People, Saving Lives

Meditec 2016





Organized First International Medical Engineering Conference where Doctors and Engineers are joining together to work on Healthcare issues

IBRO APRC 2017 & 2019





Organized First International Neuroscience Conference where neurologist, Neurosurgeon and Biomedical Engineers are joining together to work on neurological disorder, neuromodulation and neural interventions. Also organized first Deep Brain Surgery in Bangladesh at NINS.

First DBS Surgery in Bangladesh (2017)

Deep Brain Stimulations (DBS)



Prof. Tipu Z. Aziz
PROFESSOR OF NEUROSURGERY,
University of Oxford, UK
My PhD Co-Supervisor



After success DBS Surgery


সময়




icmr
INDIAN COUNCIL OF MEDICAL RESEARCH
Symposium on
Artificial Intelligence
for Health


icmr
INDIAN COUNCIL OF MEDICAL RESEARCH
SYMPOSIUM ON
ARTIFICIAL INTELLIGENCE FOR HEALTH
Date: 12 November, 2019
In Partnership with
World Health Organization
ITU




7th ITU / WHO
FOCUS GROUP MEETING ON ARTIFICIAL INTELLIGENCE FOR HEALTH
12th-15th November, 2019
Organized By
National Institute of Communication Finance (NICF)
in
Partnership with ICMR
at

Award & Recognition



Bangladeshi Startup CMED Health of AIMS lab of UIU Won the Innovation Prize at SEEDSTARS Summit 2018

Award & Recognition





Category: SDG Inclusion
Bangladesh Innovation Award 2018

Winner: The HEAD Foundation
Innovative Social Enterprise Award 2018
DBS-NUS Social Venture Challenge Asia

$SG20,000 In Cash Prize

Recognitions

(National ICT Award [Health, Inclusion, R&D, Mobile App)



CMED Health



Autism Barta



Autism Barta



Bolte Chai

Recognitions

(National ICT Award [Health, Inclusion, R&D, Mobile App)









Recognitions

(Asia Pacific ICT Award and Global Innovation Award)

Autism Barta

Bolte Chai

CMED Health





seedstarsworld

Global Innovation Winner 2018

Dhaka Winner 2017

Acknowledgement

University of Toronto, Canada	Professor Tom Chau, Dr. Catriona Steele
Holland Bloorview Kids Rehabilitation Hospital	Dr. Evdokia Anagnostou, Dr. Jessica Brian
Chinese Academy of Sciences, China	Professor Shouyan Wang
University of Southampton, UK	Professor Mark E Lutman
Imperial College London, UK	Dr. Ravi Vaidyanathan, Dr. Michael Mace
University of Oxford, UK	Professor Tipu Aziz, Professor John Stein
Florida International University, USA	Dr. Hai Deng
King Saud University, Saudi Arabia	Dr. Ghulam Muhammad
United International University, Bangladesh	Professor Mohammad Nurul Huda

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